INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING: PUMPER, 1000 GPM; LADDER/WATER TOWER COMBINATION

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 1000-gallon-per-minute (gpm) firefighting truck, with a minimum 100-foot (ft)-long aerial ladder/water tower combination.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

RR-C-271	- Chains and Attachments, Welded and Weldless.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859	- Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-W-5044	- Walkway, Compound, Nonslip and Walkway Matting, Nonslip.
MIL-W-5050	- Walkway, Coating and Matting, Nonslip Aircraft, Application of.
MIL-F-8815/23	- Filter Element, Noncleanable, Fluid Pressure, Hydraulic Line, 3000 Psi, 5 Micron Absolute, -65 Deg. to +275 Deg. F, F14 and F18 Aircrafts, 84 GPM.
MIL-F-10062	- Engine, Preparation for Shipment and Storage of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-in.
MIL-F-24385	- Fire Extinguishing Agent, Aqueous Film Forming Foam, (AFFF), Liquid Concentrate, for Fresh and Sea Water.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-H-52471	- Hose and Hose Assemblies, Rubber, Hydraulic Pressure Type, General Specification for.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors.

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.

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MIL-STD-281	- Automobile, Trucks, Truck Tractors and Trailer Dollies, Preservation and Packaging of.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1410	- Methods for Selection of Individual Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF AGRICULTURE (DOA)

Forest Service Standard 5100-1 - Spark Arrestors.

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 Flair Drive, El Monte, CA 91731.)

(Copies of specifications, standards, handbooks, documents and publications acquired by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1	- Welding Code Structural - Steel.
ANSI/AWS D1.2	- Welding Code Structural - Aluminum.
ANSI B93.28	- Calibration of Liquid Automatic Particle Counters Using "AC" Fine Test Dust.
ANSI Z26.1	- Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 153	- Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A 240	- Plate, Sheet, and Strip for Pressure Vessels, Heat Resisting Chromium and Chromium-nickel Stainless Steel.
A 525	- Sheet Steel, Zinc-Coated (Galvanized) by the Hot-Dip Process, General Requirements.
A 380	- Cleaning and Descaling Stainless Steel Plate Equipment and Systems.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Std. No. 114	- Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 East Ninth Street, Cleveland, OH 44114.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1901	- Standard for Automotive Fire Apparatus.
NFPA 1931	- Standard for Ladders.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

Tire and Rim Association Yearbook, with Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL Standard 822 - Guide for Certification of Fire Department Pumpers.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in, or through, libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of the specification shall take precedence. Nothing in the specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The firefighting truck, hereinafter referred to as "truck", shall be a diesel-powered unit capable of pumping 1000 gpm of water, and carrying not less than 300 usable gallons (gal) of water and 30 usable gal of foam liquid with necessary equipment and passengers, to the scene of a fire at highway speeds.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any change or deviations of the truck from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 4.1.2). All components furnished on the truck that are normally considered by the Fire Service to be chromium plated or stainless steel as furnished on the contractor's commercial model truck shall be chrome plated or polished stainless steel. Equipment shall include but not limited to the following:

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a. Handrails and grabrails.
b. Hose rollers.
c. External hose connections and caps.
d. Air horns.
e. Lights and mounting base.

3.3.1 Material deterioration and control. The truck shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage environments to which the truck may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the truck may be newly fabricated from recovered materials to the maximum extent practicable, provided the truck produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the truck.

3.4 Design. The design of the truck shall be as specified in 3.4.1 through 3.4.2.11.

3.4.1 Angles of approach and departure. The design of the truck shall provide angles of approach and departure which are not less than those specified in the National Fire Protection Association (NFPA) Code 1901.

3.4.2 Major truck units. The truck design shall provide a vehicle having a chassis, cab, water tank, foam system, fire pump system, body with integral stowage units, electrical system, audible warning system, handrails and grabrails, stowed components, and an aerial ladder/water tower combination.

3.4.2.1 Chassis design. The truck design shall provide a chassis of the cab-ahead-of-engine type. The chassis shall be as specified in 3.11.1 through 3.11.1.10.2.

3.4.2.2 Cab design. The truck design shall provide a four- door cab that is a combination driver's compartment and a jump-seat (crew) area located as a cab-ahead-of-engine installation. The cab shall be as specified in 3.11.2 through 3.11.2.3.

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3.4.2.3 Water tank design. The truck design shall provide a removable water tank located aft of the fire pump system (see 3.4.2.5). The water tank shall be as specified in 3.11.3 through 3.11.3.4.

3.4.2.4 Foam system design. The truck design shall provide for the inclusion of a mechanical foam system. The foam system shall be as specified in 3.11.4 through 3.11.4.4.

3.4.2.5 Fire pump system design. The truck design shall provide for the inclusion of a fire pump system. The fire pump system shall be as specified in 3.11.5 through 3.11.5.3.6.

3.4.2.6 Truck body design. The truck design shall provide for a truck body that is a composite formed of integral stowage units, supporting members, and necessary panels. The truck body shall be as specified in 3.11.6 through 3.11.6.4.

3.4.2.7 Electrical system design. The truck design shall provide for a 12-volt, direct current (vdc) electrical system with provisions for a 110-volt, alternating current (vac) invertor system. The electrical system shall be as specified in 3.11.7 through 3.11.7.5.3.

3.4.2.8 Audible warning system design. The truck design shall provide for an audible warning system. The audible warning system shall be as specified in 3.11.8 through 3.11.8.4.

3.4.2.9 Handrail and grabrail design. The truck design shall provide for handrails and grabrails to assist personnel boarding and riding the truck. Handrails and grabrails shall be as specified in 3.11.9.

3.4.2.10 Stowed component mounting design. The truck design shall provide for the mounting of holders, brackets or mounting pads required to stow various components needed for the truck. The stowed components shall be as specified in 3.11.10 through 3.11.10.8.

3.4.2.11 Aerial ladder/water tower design. The truck design shall provide for an aerial ladder/water tower. The aerial ladder/water tower shall be as specified in 3.11.11 through 3.11.11.11.

3.5 Construction. The truck shall be constructed so that all mating surfaces are securely fastened, and reinforced where necessary, to resist the twisting and bending forces applicable to emergency fire service.

3.5.1 Interchangeability. All truck parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or by adjoining parts, except for adjustment, and without selection for fit or performance.

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3.5.2 Threaded connections. All threaded connections of the piping for the water tank, foam system, fire pump system, and aerial ladder/water tower shall conform to FED-STD-H28.

3.5.3 Wiring. All wiring, except for ignition harness, shall be color coded or numbered for identification. Functions imprinted on individual wires at intervals are acceptable. Wiring identification shall be consistent throughout the truck. Wiring shall be neatly run, with all excess wire removed. Wiring shall be protected by looms or grommets at all points of potential abrasion. Wiring shall be secured to the truck by corrosion-resistant clamps and fasteners. Wiring shall be secured in compact harnesses and attached to the truck at close intervals. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or components that require removal during maintenance, except for actual connection to components. Each 12-vdc electrical circuit shall be protected with a thermal or automatic-reset circuit breaker in accordance with SAE J553.

3.5.4 Mountings. Mountings for holders, brackets, mounting pads, mounted components (lights, horn, etc.) and for handrails and grabrails shall be constructed by a method that includes the fasteners extending entirely through the mounting surface, and a spacer. The area of the mounting surface and the spacer shall be equal to or exceed the area of the mounting surface of the device being mounted. Mountings for attached cabinets shall incorporate reinforcement that transfers load stress to body support members. Mountings, fasteners, and spacers shall be corrosion-resistant metal or hot-dipped galvanized in accordance with ASTM A 153.

3.5.5 Attaching chains. When components such as caps or plugs for inlets, outlets, or connections are specified herein as chain attached to the truck, the attaching chain shall be in accordance with RR-C-271, type II, class 6, either corrosion-resistant metal or zinc coated. The chain shall be secured to the truck by clevis attachment to a mounting pad and by clevis and swivel attachment to the cap or plug. The breaking strength of the clevises and swivels shall be not less than the breaking strength of the chain. The clevises and mounting pads shall be corrosion resistant metal or hot-dipped galvanized in accordance with ASTM A 153.

3.6 Maintainability. Provision shall be made for adjustment, servicing, or replacement of parts and components of the truck. All major components, such as the water and foam-liquid tanks, fire pump, transmission, foam proportioner, and aerial ladder/ water tower, shall be individually removable and replaceable without disassembly or removal of any other component of the truck. When access openings are used to facilitate adjustment, servicing, or replacement of parts and components, the dimensions of hand or arm access shall be in accordance with SAE J925 for arctic clothing. The edges of access openings shall be rounded and smooth to prevent personal injury. Hinged covers or removable access plates shall be provided for any access opening that will admit water, dust, or any other deleterious material during operations. Hinged covers shall be provided with quick-disconnect, positive locks and access cover plates shall be provided

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with quick-disconnect fasteners. Scheduled maintenance requirements shall be not more than 50 hours of engine running time, except for before-and-after operation servicing. Each operation in the following maintenance operation list shall be accomplished by one man, within the time specified:

a. Remove, replace, and adjust all engine-driven belts - 3/4 hour.
b. Remove and replace alternator - 1/2 hour.
c. Remove and replace engine coolant-system hoses - 1 hour.
d. Drain engine lubricating oil, remove and replace oil filter element(s), and refill crankcase - 1 hour.
e. Remove and replace fuel filter elements - 1/2 hour.
f. Disconnect battery cables, remove and replace batteries, and reconnect battery cables - 1/2 hour.
g. Drain torque converter oil and transmission oil, remove and replace all transmission filter elements and strainers, and refill torque converter and transmission- 1 hour.
h. Remove and replace starter - 1 hour.
i. Remove and replace all hydraulic system filters, screens, and strainers - 1 hour.

3.7 Performance. The truck shall perform as specified in 3.7.1 and 3.7.2.

3.7.1 Mobility. The truck shall conform to all performance and mobility requirements of chapter 11, NFPA Code No. 1901, when loaded as follows:

a. Water tank filled with 200 gal of water.
b. Foam-liquid tank filled with 30 gal of foam liquid.
c. Fuel tank filled with diesel fuel.
d. All lubricant and hydraulic reservoirs filled to recommended capacity with appropriate fluids.
e. An evenly distributed, simulated load of 1500 pounds (lb).
f. A crew of five, each weighing 175 lbs, or an equivalent deadweight substitute for any of them.

3.7.2 Pumping capacity. Each truck shall be certified by UL as capable of delivering 1000 gpm of water, in an ambient temperature (see 6.4.1) of +125 °F, at any altitude from sea level to 2000 feet (ft) above sea level. For the purpose of this requirement, net pump pressure shall be calculated in accordance with NFPA Code No. 1901.

3.8 Human factors. The truck shall conform to MIL-STD-1472 criteria, as applicable, and also conform to the human factors requirements specified in 3.8.1 through 3.8.2.2.

3.8.1 Operator compartment. The operator characteristics of the driver's compartment (see 3.11.2.1), the pump operator's panel (see 3.11.6.3), and the control station for the aerial ladder/water tower (see 3.11.11.7), shall provide for operation by personnel ranging from the small person clothed through the large person arctic clothed, in accordance with SAE J833 and SAE J925.

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3.8.2 Labeling and marking.

3.8.2.1 Control labels. The control labels for gages, instruments, switches and control handles shall be in accordance with the labeling requirements of MIL-STD-1472. Universal symbology shall be applied to driver's gages, instruments and controls on the dash in accordance with SAE J1500. All operator contacted valves shall be labelled with their function (i.e., foam discharge, water discharge, etc.), with "open-close" and with double-ended arrows which indicates direction of operation for open-close.

3.8.2.2 Identification and instruction plates. The identification plate (see 3.18) and instruction plates required shall be located on the truck on the flattest, most uncluttered surface available, in such a position that they are not obscured by adjacent items, so as to minimize obscurement by grease, grime, and dirt, and to preclude accidental removal, obstruction or handling damage. Identification and instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, size as applicable. All plates shall be attached by screws, bolts, or rivets.

3.8.2.3 Nameplate marking. A nameplate equal in height to not less than 80 percent of the depth of the base ladder section and not less than 5 feet long shall be attached to both sides of the base section of the ladder. The plate shall be centered top to bottom of the base section and shall be lettered in block-style letters "U.S. Army" in the largest-size lettering practical for the size nameplate used. Lettering shall be gloss-black against a gloss-white background.

3.9 Safety. The truck shall conform to the safety requirements specified in 3.9.1 through 3.9.3.

3.9.1 Noise limits. The noise produced by the truck shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2 and 5.4, when tested in accordance with 4.5.2.4. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit at an operator's position is obtained from the procuring activity. If permission to exceed 85 dB(A) at an operator's position is obtained, then noise hazard signs shall be posted at the position. The sign shall conform to MIL-STD-1474, 4.3, except the sign shall state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET" and shall be readable at the distance indicated on the sign. The distance shall be the maximum distance obtained from an 85 dB(A) noise contour conducted in accordance with 4.5.2.4.

3.9.2 Personnel protection. All rotating or reciprocating parts and other parts that are of such nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating efficiency of the truck.

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3.9.3 Boarding and working protection. All walking surfaces and treadplates shall be provided an antiskid surface which conforms to MIL-W-5044 and is applied in accordance with MIL-W-5050. All platforms, steps, and handholds shall permit unhampered and nonhazardous boarding, working, and departure from the truck. Hose rollers shall not serve as handrails or grabrails.

3.10 Environmental requirements.

3.10.1 Altitude pumping. When specified (see 6.2), the truck shall be capable of delivering its rated pumping capacity at the specified service altitude in an ambient temperature of 75 °F ±10 °F. Net pump pressure shall be calculated in accordance with UL Standard 822. The contractor may install modification kit(s) for truck service at any altitude above 2000 ft above sea level to conform to this requirement.

3.10.2 Cold starting and operating. The truck shall be capable of starting and operating at -25 °F, after the truck has been cold-soaked for not less than 24 hours at an ambient temperature of -25 °F. A built-in, ether-priming, cold-starting aid may be installed to conform to this requirement. The cold- starting aid, if installed, shall conform to MIL-E-52649.

3.10.3 Hot starting. The truck shall be capable of starting and operating at not less than 125 °F ambient air temperature when the temperature of the engine coolant is at the engine manufacturer's recommended operating temperature.

3.11 Detailed requirements.

3.11.1 Chassis. The truck shall have a custom, front-wheel-steer, tandem, rear-wheel-drive-type chassis.

3.11.1.1 Frame. The chassis frame shall be a heavy-duty design for fire service use and shall have crossmembers, exclusive of engine supports, to support the gross weight of the cab, body, powerplant, pump, water tank, all other equipment and accessories, and the load specified in 3.7.1, without evidence of distortion that will adversely affect operations. No alteration of the frame shall be made which will reduce its strength.

3.11.1.2 Engine. For the purpose of selecting the engine, the truck shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, except as otherwise specified herein. All engine accessories shall be furnished or approved by the engine manufacturer for fire service application. The engine shall conform to all operational requirements and conditions specified in table I of MIL-STD-1410, or the engine manufacturer's recommended operating limits, when operating in a +125 °F ambient air temperature. The engine shall be a liquid-cooled diesel type and shall have not less than eight cylinders with replaceable liners. The engine shall be furnished with a governor that is set to limit engine speed to the revolutions per minute (rpm) recommended by the engine manufacturer.

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3.11.1.2.1 Engine horsepower. The engine shall develop not less than 400 brake horsepower (bhp), at an engine speed of not more than 2600 rpm, when operating on diesel fuel conforming to VV-F-800 and lubricating oil conforming to MIL-L-2104 and MIL-L-46167. The power and speed rating of the engine shall be such, that when the truck is operated as specified herein, it shall require not more than the Environmental Protection Agency (EPA) certified horsepower for that engine.

3.11.1.2.1.1 BHP certification. The contractor shall obtain and have available for review by the contracting officer or designated representative a net bhp certification for the engine provided in the truck.

3.11.1.2.2 Engine lubrication. The engine lubrication system shall be the engine manufacturer's standard type and shall include an oil filter. Only heavy-duty, full-flow oil filter(s) shall be utilized in the engine lubrication system, and shall be accessible without removing other components. When applicable, oil lines to and from the oil filter(s) shall be installed to minimize the effect of vibration and shall be of flexible hose in accordance with the requirements of SAE J30(d), with reusable fittings. A tag shall be attached to the transmission shift lever identifying type of lubrication used.

3.11.1.2.3 Engine fuel system. The capacity of the engine fuel tank shall be sufficient to operate the fire pump at rated capacity for not less than 4 hours without refueling, but shall not be less than 50 gal. The fuel tank shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust system, or electrical equipment. The fuel system filler opening shall permit refueling from the spout of a military 5-gal container in accordance with MIL-C-1283. Fuel lines shall be adequately supported and shall be protected against mechanical damage. Construction and mounting shall be such that neither vibration nor tightening of mounting devices will cause leaks to develop. The tank shall have a drain which is located so that tank contents can be drained into a vessel. The bottom of the fuel tank and its supporting members shall not be lower than the lowest adjacent part of the chassis or body. The filler cap shall not protrude beyond the body. The tank shall be labelled "Diesel Fuel Only" in 0.5-inch-high white letters near the filler cap.

3.11.1.2.4 Engine cooling system. The engine cooling system shall be of the capacity recommended by the engine manufacturer. Draincocks shall be provided at the low points in the cooling system and at such other points necessary to drain the cooling system. The draincocks shall be of the hand-operated type that will remain closed under the most severe operating conditions encountered by the truck. The engine cooling system shall be equipped with a supplemental heat exchanger. The heat exchanger shall admit water from the fire pump discharge manifold into the heat-exchanger jacket for cooling engine coolant without admixture. Engine coolant hoses shall conform to SAE J20, heavy-duty type. Engine coolant hose clamps shall be of the double- wrap, stainless-steel type in accordance with SAE J536, type F, style 1, 2 or 4. Gaskets, seals, and packing used in the cooling system shall be manufactured of materials capable of resisting deterioration from exposure to mixtures of water and ethyleneglycol-based antifreeze conforming to MIL-A-46153.

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3.11.1.2.5 Engine starting system. The engine starting system shall prevent engine start unless the transmission is in neutral. The ignition system shall be equipped with a keyless switch and a pilot light.

3.11.1.2.6 Engine exhaust system. The exhaust system shall include a muffler and tailpipe. The exhaust system shall be located so that entrance of exhaust gases into the driver and crew area compartment shall be minimized under all conditions of operation. Exhaust and other discharges from the truck shall be directed so that they shall not endanger personnel or obstruct the view of the driver or fire pump operator. Location of the tailpipe shall prevent heating of tires, disturbance of road dust, and any clogging by mud or water. The muffler shall be located not closer than 2 ft from the engine fuel tank. Where this is not possible, fuel shielding shall be used to protect the engine fuel tank from any part of the exhaust system, and insulation and shielding elsewhere. The exhaust system shall be provided with a spark arrestor conforming to Forest Service Standard 5100-1 and SAE J350 for not less than 80-percent efficiency. The exhaust system shall not constitute a personnel contact hazard.

3.11.1.3 Drive assembly. The entire drive assembly between the engine and the drive wheels shall have a torque capacity that is not less than the maximum torque developed by the engine through all intervening speed and torque ratios.

3.11.1.3.1 Transmission. The transmission shall be the fully automatic type (see 6.4.2), with not less than a four forward and one reverse gear. The transmission shall be rated for heavy-duty fire service and designed to match the engine torque-to-load demand. The transmission shall be protected so that the driver cannot inadvertently shift into reverse. When the transmission is used to drive the fire pump or hydraulic pumps (see 3.11.11.1.3), a manually operated locking device shall ensure that the transmission and its shift lever remains in place. When power transfer through a power-takeoff (PTO) is engaged, a red warning light shall be illuminated on the driver's dashboard (see 3.11.2.1.2).

3.11.1.3.2 Drive axle. The drive axle shall be of the full-floating type.

3.11.1.4 Suspension system.

3.11.1.4.1 Springs. Front and rear springs shall sustain the distributed gross vehicle weight (gvw) without evidence of overload or permanent set. The springs shall be designed and suspended for the superimposed load so as to absorb shock and limit sway. Individual wheel loadings, with the truck loaded as specified in 3.7.1, shall be not greater than the rated capacities of the respective springs measured at the ground.

3.11.1.4.2 Shock absorbers. Shock absorbers shall be the double-acting, hydraulic type, which are matched to the suspension capacity, and are suitable for the gross axle weight rating (gawr). Shock absorbers shall be installed on front axles if required.

3.11.1.5 Wheels and tires. Wheels and tires shall conform to the Tire and Rim Association (TRA) requirements for 50 to 60 mile per hour (mph) speeds for vehicles of the truck's gvw. A quick-disconnect type air chuck shall be installed on the lower left corner of the pump panel to provide an air supply to

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inflate the truck tires. A 24-foot air hose, with mating connection on one end to connect to the air chuck and a device to inflate the tire on the other end, shall be furnished with each truck. Air shall be provided from the air-brake system. Fenderetts or crowns used to reduce tire splash shall be chrome plated.

3.11.1.5.1 Front wheels. The front wheels shall be ten-hole disk wheels. Wheels shall be in balance.

3.11.1.5.2 Rear wheels. The rear wheels shall be dual, ten- hole disk wheels. Wheels shall be in balance. Clearance in the wheel wells shall be provided for the use of dual-type snow chains.

3.11.1.5.3 Tires. Tires shall be not less than 100-level quality and shall not be more than 12 months old on the date of shipment to the Government. Front tires shall be sized for the load imposed and shall have a highway tread. Rear tires shall be sized for the load imposed and shall have a traction lug tread.

3.11.1.5.3.1 Tire valves, caps, and cores. Tire valves shall clear both wheels and brakedrums. The valves shall be furnished with caps and heat-resistant TRA G-4 cores. Tire pressures recommended by the TRA Yearbook shall be stencilled in 0.5-inch- high white letters above and near the tires. These stencils shall read "TP (specify) PSI".

3.11.1.5.4 Spare. A spare wheel and tire assembly shall be furnished and shipped as an overpack.

3.11.1.6 Brakes. The truck shall incorporate a straight air-brake system. All brakes shall conform to the latest Department of Transportation (DOT) Motor Carrier Safety Regulation requirements and the provisions of 3.11.1.6.1 through 3.11.1.6.3, and shall meet the performance requirements of SAE J9926 for the size vehicle furnished.

3.11.1.6.1 Service brakes. The service brakes shall control and hold the truck, when loaded as specified in 3.7.1, on a 30- percent grade. The service brake shall stop the truck within 30 ft, when loaded as specified in 3.7.1, from a speed of 20 miles per hour (mph) on dry, hard, level surfaces, free of loose material. The service-brake mechanisms shall be accessible for exterior adjustment. Brakelines shall be anchored to the chassis and protected from damage. Provisions to prevent system freezing shall be incorporated in the service-brake system.

3.11.1.6.2 Parking brake. The parking brake shall be all-wheel locking. The parking brake shall be the air-release type. The parking brake air system, when charged to not less than 100 lbs per square inch (psi), shall not leak more than 25 percent in 24 hours. The parking brake control shall be of the "push-pull" type. The parking-brake system shall be provided with an emergency brake-release mechanism. Rear parking brakes shall be provided with spring loaded brake.

3.11.1.6.3 Engine brake. The engine brake shall be an engine exhaust-brake system designed for the engine used in the truck. Braking shall be accomplished by motion transfer through a master-slave piston arrangement which opens the cylinder exhaust valves near the top of the normal compression stroke, releasing

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the compressed cylinder charge to exhaust. The engine-brake control shall be an "on-off" switch, which is labeled "ENGINE BRAKE" and "ON, OFF" as appropriate, which permits the engine brake system to be energized by removing pressure from the accelerator pedal when the switch is in the "on" position. The design shall provide that the brake lights shall come on when the engine brake is on.

3.11.1.7 Generating system. The truck shall be equipped with a diesel engine-driven, 110-Vac system. The system shall have an output rating of not less than 300 Wac. The system shall be sized to provide not less than 3000 watts (W) of 110 Vac from the system (see 3.11.2.1.7) when the engine is operating at the engine manufacturer's recommended speed.

3.11.1.8 Steering system. The truck shall be equipped with a power-assisted, hydraulic-type, steering mechanism, which shall require not more than 15-lbs turning force to steer the rolling vehicle under normal operation or 50-lb force under power-assist-failure operation. The steering mechanism shall preserve, to the maximum extent practicable, the features of directional stability inherent in the manually operated steering mechanism and shall be effective at all vehicle speeds. The mechanism shall permit manual steering in the event of failure of the hydraulic system. The steering mechanism shall include a means for rapid adjustment of lost motion. The mechanism shall provide a cramping angle of the front wheels of not less than 28 degrees right and left at the wheel on the inside of the turning circle. The steering wheel shall be mounted on a rigid and sturdy column. The steering wheel shall be ample size, but shall not interfere with any other projecting equipment in the driver's compartment or operation of the vehicle.

3.11.1.9 Steps and bumpers.

3.11.1.9.1 Rear step. The truck shall be equipped with steps or ladders and hand rails to assist personnel in climbing to the ladder turn table and assist in packing and removing hose. Fold or pull down steps shall be provided when the distance between the ground and the loaded truck is more than 25.5 inches. Steps shall be as specified in 3.17.

3.11.1.9.2 Front bumper. The front bumper shall be a heavy duty unit attached directly to the frame. The bumper shall be chromium plated or polished stainless steel.

3.11.1.10 Towing hooks and towing lugs. Towing hooks and towing lugs shall be provided on the truck. The location of the attachment shall not reduce the angle of departure.

3.11.1.10.1 Towing hooks. Rear mounted towing hooks or eyes shall be installed symmetrically about the center line of the truck. Towing eyes shall not be less than 3 inches in diameter. The towing devices shall be attached to the frame.

3.11.1.10.2 Towing lugs. Two vehicular towing lugs shall be installed on the front of the truck. The towing lugs shall be attached directly to the frame.

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3.11.2 Cab. The truck shall have a 4 door cab and shall have a closed driver's compartment forward and a jump-seat area aft. The partition between the driver's compartment and the jump-seat area shall have a sliding glass window. Front wheelhousings of the cab shall be adequate to permit steering as specified herein. A removable section shall be provided in the cab roof to provide ease in removal of the truck engine. The cab shall conform to the heating, ventilating, requirements of SAE J1129. Windows shall be provided over the driver and the passenger to allow visibility of the highest point of the ladder when the ladder is in the road position.

3.11.2.1 Driver's compartment. The driver's compartment shall be separated from the engine by an insulated firewall. The interior of the compartment shall have a headliner or false roof and finish trim on exposed vertical panels. All wiring and tubing to exterior roof-mounted lights and horns, interior lights, and other accessories shall be run behind the headliner and vertical panels. The compartment interior shall be fitted with insulation behind the headliner as well as vertical panels to retard heat loss and noise intrusion. All compartment openings shall be sealed with rubber boots, elastomer grommets, elastomer seals, weatherstripping, or body compound, as appropriate.

3.11.2.1.1 Driver compartment seats. There shall be two seats provided in the driver compartment. The seats shall have upholstered seats and backs. The cushion material shall be foam rubber or polyurethane, slip covered with coated vinyl upholstery having a minimum weight of 45 ounces (oz) per linear yard. The driver's seat shall be adjustable fore and aft and up and down in a manner that assures that 3.8.1 and SAE J833 and J925 requirements are met. The passenger seat shall be fixed and shall have a seat back to hold a 30-minute air pack, as specified in 3.11.2.2.2. The seat shall conform to dimensions specified in 3.11.2.2.2 and SAE J833 for the 50th percentile male. The driver's minimum seat dimensions shall conform to SAE J899.

3.11.2.1.1.1 Jump seat (crew) area. The crew area shall provided with seating for not less than four firefighters. There shall be two seats facing aft with padded backrests to hold 30 minute air packs and two fold down seats facing front. There shall be one each seat on either side of the engine cover. Seat material and seat belts shall be as specified in 3.11.2.1.1.

3.11.2.1.2 Driver's dashboard. The driver's dashboard shall extend across the full width of the driver's compartment. The dashboard shall be installed at such an angle, or provided with an upper shroud, so that any dash lighting shall not reflect in the windshield, or shine directly in the driver or passenger's eyes. Depth of the dashboard shall be sufficient to mount all of the gages, instruments, switches, monitor lights, controls, and compartments specified herein. All of the components shall be directly visible to the driver when driving the truck.

3.11.2.1.2.1 Driver's gages and instruments. The driver's gages and instruments required to monitor truck operation and performance shall be in accordance with SAE J209 and shall be installed in the dashboard. The installation of the gages and instruments shall facilitate removal and servicing. The gages and instruments for monitoring truck operation and performance shall include as a minimum, but shall not be limited to, the following:

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a. Speedometer with recording odometer.
b. Tachometer with 0 to 4000 rpm range.
c. Engine oil pressure gage.
d. Brake air-tank pressure gage.
e. Engine coolant temperature gage.
f. Transmission oil temperature gage.
g. Voltmeter.
h. Fuel gage.
i. Engine hourmeter which indicates actual engine operating hours up to 9999.9 hours in tenths of an hour.

The gages and instruments shall be individually illuminated for night operation. Actuation of nightlighting shall be controlled by the headlight switch. The illumination of driver's gages/instruments shall be rheostat controlled and provide a display luminance of 0.1 to 3.5 candela per meter^2 (cd/m^2).

3.11.2.1.2.2 Driver's operating controls. The driver's operating controls shall be installed in the driver's compartment in accordance with the requirements of MIL-STD-1472 for controls. The following controls shall be installed in accordance with SAE J1138 and SAE J898:

a. Accelerator pedal, transmission shift-control, and service-brake pedal on the floor.
b. Ignition switch, parking-brake control, engine-brake switch and control for parking-brake emergency-release mechanism on the dashboard.
c. Lighting switches for truck operating lights (see 3.11.7.4.1) and equipment operating lights (see 3.11.7.4.2) requiring driver control on the dashboard.

3.11.2.1.2.3 Gage and instrument monitors. Remote indicating lights shall be furnished to provide driver warning of the conditions specified in a. through d. below. The indicating lights shall be red and shall light when the condition specified occurs. Means shall be provided to check operation of the lights. In addition, an audible alarm that exceeds operating noise levels by 20 dB(A) shall sound when the condition specified in d and e occurs.

a. When the engine is operating and the alternator is not charging.
b. When the engine is operating and the oil pressure is below the engine manufacturer's recommended minimum.
c. When the engine is operating and the engine coolant temperature is above the engine manufacturer's recommended maximum.
d. When the engine is operating and the transmission fluid temperature is above the transmission manufacturer's recommended maximum.
e. When the air pressure available is less than required for proper brake operation, the lamp shall flash at three to five flashes per second.

3.11.2.1.3 Glass and glazing. Safety glazing of the windshield, door windows, side windows, and rear window shall be provided. Glazing shall conform to ANSI Z26.1, type AS1 for the windshield and type AS1 or AS2 for other windows. Glass

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shall be tinted. Fixed or hinged glazing shall be sealed in waterproof, stress-absorbent moldings. Movable windows shall be sealed in extra-heavy-duty, pile-lined channels. The cab shall be furnished with the maximum window area practical for the cab furnished.

3.11.2.1.4 Windshield wiping and cleaning system. Windshield wipers shall be mounted to facilitate clear vision in accordance with SAE J198. Wiper assemblies shall include means of wiper speed control and return of wiper blades and arms to a retracted position out of driver vision when not in use. The windshield cleaning system shall include a reservoir and a device that will spray the path of the wiper blades with water or cleansing solution. The reservoir shall have a capacity of not less than 2 quarts. The windshield wiping and cleaning system shall conform to SAE J198.

3.11.2.1.5 Heater. A recirculating hot-air heater, which uses engine coolant as a heat source, shall be installed in the driver's compartment. The heater discharge outlets and ducts shall direct heated air to the compartment floor and to windshield defroster louvers in the top of the dashboard, in accordance with SAE J1129 and SAE J382. The heater shall be equipped with temperature, blower, and duct controls that are dashboard or console mounted.

3.11.2.1.6 Transmission shift-control pedestal. The transmission shift-control lever shall be mounted on a pedestal to the right of the driver seat or on the dashboard. A positive indicator for each gear or range selected shall be incorporated. The transmission shall be protected where the operator cannot inadvertently shift into reverse. An electrically operated back-up warning system conforming to SAE J994 shall be installed to automatically operate when the transmission is put in the reverse mode.

3.11.2.1.7 Inverter. A power supply of not less than 3000 W 110 Vac 60 hertz shall be furnished and mounted in the storage space under the bedded ladder. The system shall be a diesel powered, air cooled system with start and stop controls located in the cab and at the fire pump panel. The system shall be provided with a voltage and ampere meter at both locations. The system shall be protected by not less than an overheat and low oil pressure automatic safety stop controls. Fuel for the unit shall be supplied from the fire truck fuel tank.

3.11.2.1.8 Communications compartment. There shall be a space provided in the driver compartment for installation of communication equipment. The communication compartment space shall be located in a position that will permit either driver or front passenger operation of the communication equipment. The size of the communication space shall be not less than 1.5 ft wide by 1.0 ft high by 1.5 ft deep. Color coded (black and red) wiring, which provides 12 Vdc at 30 amps, shall be run to the communications compartment.

3.11.2.1.9 Driver compartment accessories. The driver compartment shall be equipped with the accessories specified in 3.11.2.1.9.1 through 3.11.2.1.9.4.

3.11.2.1.9.1 Rearview mirrors. West-coast-type outside rear-view mirrors shall be mounted on each side of the cab. The mirrors shall be combination straight and convex type. Stick-on convex mirrors are not acceptable. The

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mirrors shall be mounted on foldback support arms and in a position that will allow the driver to survey both sides of the truck. Mirror vision shall not be impaired by truck operation.

3.11.2.1.9.2 Seatbelts. Adjustable, weather-resistant, safety seatbelts, in accordance with SAE J117, type I, with seatbelt anchors in accordance with SAE J787, shall be furnished for all passengers. The seatbelts shall have noncorrosive, metal-to-metal, quick-release buckles and seatbelt retractors.

3.11.2.1.9.3 Interior lights. Interior lighting in the driver compartment shall be provided by a dome light mounted in the headliner or false roof and cab step-lights. The dome light shall have an "on-off" control switch (see 3.11.7.5.1). The lights shall be wired to door switches; when either door is opened, the lights will illuminate.

3.11.2.1.9.4 Sun visors. Dual sun visors shall be provided in the driver's compartment for the driver and front passenger.

3.11.2.2 Jump-seat area. The jump-seat area of the cab shall be equipped with four jump seats, two stationary, and two fold-down separated by the engine cover. The canopy over the jump seats shall have a removable section to accommodate engine removal. The area shall be finished as specified in 3.11.2.1. Each jump seat shall have a seatbelt in accordance with 3.11.2.1.9.2. The jump seat area shall be provided with roll down glass in the doors and sliding glass panels in all other glass areas to provide maximum ventilation. Two power roof vents shall be equally spaced over the jump seats.

3.11.2.2.1 Engine cover. The engine cover shall extend from the rear partition of the driver compartment to the front of the body. The engine cover shall be constructed of aluminum with external surfaces of four-way-pattern tread plate. Cover sides and top shall be double panel construction with heat and sound-deadening insulation sandwiched between the panels. The cover top shall be hinged at each side of the top, and the cover sides shall be removable. The engine cover may be equipped with ventilating louvers. The ventilating louvers shall direct the engine compartment air away from the occupants of the jump seats. The engine cover shall be removable for servicing the engine. Access panels shall be provided to check engine oil and coolant without removing the engine side panels.

3.11.2.2.2 Jump-seats. A jump seats shall be located on either side of the engine cover. The space for each jump seat shall be not less than 22 inches wide. A step, or steps, shall be provided for passenger access to the jump seats. The seating portion of the jump seats shall be not less than 22 inches wide. The cushion material shall be foam rubber or polyurethane, slip covered with coated vinyl upholstery having a minimum weight of 45 oz per linear yard. The back of each rear facing seat shall have a padded, upholstered, quick-release-type, adjustable bracket capable of holding a complete 30-minute breathing apparatus. With a complete 30-minute breathing apparatus mounted in the bracket, the flat usable seating area of the jump seats shall be not less than 14 inches from front to rear. Breathing apparatus shall not be furnished.

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3.11.2.2.2.1 Fold-down jump seats. The fold-down jump seat shall be located on either side of the engine cover. The seat bottom shall automatically fold up when not in use. Seats and back rests shall be upholstered as specified in 3.11.2.2.2. Brackets for a 30-minute breathing apparatus shall not be furnished for these seats. The seats shall face forward.

3.11.2.2.2.2 Dome lights. Dome lights, with an "on-off" control switch, shall be provided in the jump-seat area between each pair of seats.

3.11.2.3 Soft suction hose compartment. A metal compartment of sufficient size to stow a 12-ft-long section of 6-inch-dia soft suction hose (see 3.11.10.1) shall be installed at the front of the truck, behind the bumper and in front of the cab. The compartment interior shall be free from projecting nuts or reinforcing brackets. Exposed bolt or rivet heads in the interior of the compartment shall be of the oval type. The compartment shall be equipped with drain holes and shall be finished inside to avoid chafing of the hose. A filler plate shall be installed to cover the area between the cab and bumper not covered by the hose compartment. The plate shall hold not less than 300 pounds per square foot. One corner of the compartment shall be provided with a device to hold a hydrant wrench in the vertical position.

3.11.3 Water tank. The water tank shall be chassis-mounted aft of the fire pump (see 3.11.5.1) and independent of the truck body.

3.11.3.1 Water tank size. The width of the water tank shall be not more than the width of the large-hose bin of the body (see 3.11.6.1.2). The capacity of the water tank shall be not less than 200 usable gallons. The shape of the water tank and the position of the water tank outlet shall provide that water can be pumped from the tank when the truck is standing in any plane on a 20-percent grade as specified by NFPA 1901, 3.6.2.3. The interior of the water tank shall be baffled in accordance with NFPA Code No. 1901. The entire water tank top shall be removable. To prevent spillage, the water tank top shall be gasketed and retained in place by capscrews or similar positive fasteners that do not require special tools to tighten or remove. Fasteners shall not come in contact with tank water.

3.11.3.2 Water tank structure. The water tank, top and tank mountings shall be manufactured of welded stainless steel conforming to ASTM A 240, type S30403. After manufacture, and before installation of the water tank on the truck frame, the entire tank shall be sandblasted or subjected to a pickling treatment that will remove all mill and weld scale, and then thoroughly rinsed and cleaned to remove all debris.

3.11.3.3 Water tank fill and drain. The tank shall be equipped with a fill opening having an area of not less than 100 square inches to permit easy filling with a 3-inch fire hose coupling. The tank fill opening shall be equipped with a removable screen. The screen mesh shall have openings not larger than 0.250 inches. The closure of the fill opening shall be designed so that water shall not leak from the opening during acceleration, deceleration, or other movement of the truck from a full or partially filled water tank. The water tank shall be

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equipped with a nonsplash combination overflow drain and vent pipe. The overflow drain and vent pipe shall extend downward, toward the rear of the truck, and shall be directed inboard to the ground so that rear-wheel traction is not affected by water from the drain.

3.11.3.4 Water tank outlet. The water tank outlet shall be not less than 2.5 inch dia.

3.11.4 Foam system. The truck shall be provided with a mechanical foam system. The foam system shall be an around-the-pump type. The foam system shall consist of a foam-liquid tank, a foam-liquid metering valve or proportioner, fittings, piping, valves and screens as specified herein. The foam system shall provide for the delivery of foam-liquid to the fire pump in a volume that will produce an output of 3- to 6-percent foam solution throughout the operating pressure range of the fire pump at discharge capacities of 3000 to 1000 gpm. All parts and components of the foam system that will contact liquid foam shall be of austenitic or duplex stainless steel. Brass, zinc coated metals, or 400 series stainless steel shall not be used. If manuals are furnished with the fire truck, a complete foam system diagram shall be incorporated in the operation and maintenance manuals furnished with the truck.

3.11.4.1 Foam-liquid tank. The truck shall be equipped with a removable foam-liquid tank. The foam-liquid tank shall be mounted near the fire pump.

3.11.4.1.1 Foam-liquid tank size. The foam-liquid tank shall have a capacity of not less than 100 usable gal when filled through the opening described in 3.11.4.1.3. The interior of the tank shall be baffled to prevent excessive surging (see 6.4.3) and foaming of the liquid.

3.11.4.1.2 Foam-liquid tank structure. The foam-liquid tank shall be manufactured of welded stainless steel conforming to ASTM A 240, type S30403. After manufacture, and before installation in the truck, the entire tank shall be sandblasted or subjected to a pickling treatment that will remove all mill and weld scale, and then thoroughly rinsed and cleaned to remove all debris.

3.11.4.1.3 Foam-liquid tank filler opening. The foam-liquid tank shall have a filler opening that will allow filling from a 5-gal container. The filler opening shall have a neck that extends into the tank interior to reduce foaming overflow while filling. The filler opening shall be large enough to permit hand cleaning of the inside of the tank. The filler opening shall be equipped with a removable screen of a mesh size that will prevent entry of foreign matter into the tank. The filler opening shall be equipped with a gasketed, latching-type cover.

3.11.4.1.4 Foam-liquid tank vent. The foam-liquid tank shall have a splashproof vent that is piped to the outside of the truck.

3.11.4.1.5 Foam-liquid tank outlet. The outlet of the foam- liquid tank shall be equipped with a removable screen to prevent foreign matter or jelled foam-liquid from entering the piping and valving of the foam system.

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3.11.4.1.6 Foam-liquid tank drain. The foam-liquid tank shall be equipped with a drain outlet. A drain valve shall be installed at the outlet. A drain pipe shall be attached to the drain valve and shall be located so that the tank is drained completely and is directed downward so the discharge point is below the bottom of the truck frame and does not drain in a manner that reduces rear-wheel traction.

3.11.4.2 Foam-liquid metering valve or proportioner. A calibrated foam-liquid metering valve or proportioner shall be provided in the foam system to regulate the discharge or proportioning of foam-liquid from the foam-liquid tank to the fire pump. The foam-liquid metering valve or proportioner shall be designed for remote regulation of the foam-liquid flow as a percentage of the total solution flow from the fire pump.

3.11.4.3 Foam system piping and valves. A quick-acting, corrosion-resistant shutoff valve shall be provided in the foam system piping between the foam-liquid tank and the foam-liquid metering valve or proportioner. The shutoff valve shall be as close to the foam-liquid tank as possible. A check valve shall be provided between the foam-liquid metering valve or proportioner and the eductor at the free pump suction manifold to prevent water or solution from flowing back from the manifold to the foam-liquid delivery system.

3.11.4.3.1 Foam system flushing provisions. A means for flushing the foam piping and valving shall be installed in the foam system. The flushing provisions shall be designed to permit a complete flushing of the system and shall assure that no foam- liquid residue will accumulate to clog the system when proper flushing has been performed. If manuals are furnished with the fire truck, complete flushing and cleaning instructions shall be incorporated in the operation and maintenance manual furnished with the truck.

3.11.4.4 Foam system controls. Controls for operation of the foam system shall be located as specified in 3.11.6.3.1.3. An instruction plate showing the proper settings of the metering valve or proportioner shall be located adjacent to the foam system controls.

3.11.5 Fire pump system. The fire pump system shall incorporate a fire pump, suction section, discharge section, piping, and drains. The fire pump system shall exhibit no evidence of leaking when subjected to the hydrostatic test specified in 4.5.2.10. If manuals are furnished with the fire truck, a complete diagram of the pumping system shall be incorporated in the operation and maintenance manuals furnished with the truck.

3.11.5.1 Fire pump. The pump shall be of the centrifuged type. The pump shall have a certified Underwriters Laboratories (UL) Standard 822 minimum rating of 1000 gpm. The fire pump shall be equipped with a certified UL priming device that conforms to UL Standard 822. A UL certification plate shall be attached to the pump panel. A copy of the UL certification test report shall be a part of the operator's manual that accompanies each truck.

3.11.5.1.1 Fire pump drive. The fire pump shall be driven by the truck engine. The temperature rise in the gear oil of the pump drive shall not exceed the maximum gear-oil temperature recommended by the fire pump manufacturer when

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pumping in continuous operation. The pump drive may be equipped with supplemental oil cooling to assure that fire pump gear-oil temperature rise will not exceed the fire pump manufacturer's recommended maximum gear-oil temperature for continuous operation. The control of the fire pump drive shall be as specified in 3.11.6.3.1.7.

3.11.5.2 Suction section. The fire pump system suction section shall include a fire pump suction manifold, large suction inlets, small suction inlets, and the necessary piping, valves, and connections required to assure the fire pump will draft as specified herein. The piping for the front suction section shall be corrosion-resistant metal, passivated in accordance with ASTM A 525, G 90 coating. A UL certification plate shall be attached to the pump panel. A copy of the UL certification test report shall be a part of the operator's manual that accompanies each truck.

3.11.5.2.1 Fire pump suction manifold. The fire pump suction manifold shall be provided with the necessary ports, sized to accommodate the flow volumes indicated for piping attachments as specified herein. The fire pump suction manifold shall direct suction flows to the fire pump inlet.

3.11.5.2.2 Large suction inlets. Large suction inlets shall be 6 inches dia. There shall be three large suction inlets mounted on each truck. One inlet shall be located at the front of the truck and one inlet on each side of the truck. The inlets shall be not more than 4 feet above ground level. Each inlet shall have 6-inch National Hose (NH) male threads (thds) and machined ends for suction hose coupling or valve gasket seating. Each inlet shall be furnished with a readily accessible, removable brass strainer. The inlet at the front of the truck shall be equipped with a butterfly-type valve. The valve shall have a positive, leakproof seal, and a hand-adjustable control handle. The front suction inlet hose coupling and the valve shall not extend past the bumper. The two side inlets shall be furnished with long-handled, gasketed caps. The suction inlets shall be piped to appropriate sized ports in the suction manifold.

3.11.5.2.3 Small suction inlets. There shall be two small suction inlets, 2.5 inches dia, located on the left or streetside of the truck, at the bottom of the pump operator's panel (see 3.11.6.3). The inlets shall be gated-and-gasketed, female, swivel-suction type with internal NH thds. Each inlet shall be furnished with a rocker-lug, male plug. The plugs shall be attached to the truck with a chain. The inlets shall be piped to appropriate-sized ports in the fire pump suction manifold through 2.5-inch-dia, balanced, full flow valves. The valves shall be operated by individual locking control handles by using one-hand. Controls for the valves shall be as specified in 3.11.6.3.1.5. Austenitic or duplex stainless steel. Brass, zinc coated metals, or 400 series stainless steel shall not be used.

3.11.5.2.4 Water tank suction piping. The manufacturer's standard quick-opening shut-off valve shall be installed in the piping from the water tank to the fire pump suction manifold. The valve shall be designed for remote opening and closing. A check valve shall also be installed in the piping to prevent any backflow of water or foam liquid solution from the fire pump from reaching the shutoff valve or the water tank. The piping between the shutoff valve and the fire pump suction manifold may incorporate flexible, noncollapsible connections

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to absorb vibration. The piping shall have a drain valve of not less than 0.75-inch dia installed at its lowest point to permit draining of the piping and check valve while the shutoff valve is closed. The drain valve shall be designed for remote control. The check valve shall have a small relief hole to avoid high pump-pressure buildup when all valves are closed. A drain pipe from the drain valve shall discharge to a point below the bottom of the frame. The piping, valves, and connections shall be sized to flow not less than 200 gpm and shall withstand a working pressure of not less than 120 psig. The controls for the shutoff valve and the drain valve shall be as specified in 3.11.6.3.1.4.

3.11.5.2.5 Suction manifold fittings. The fittings that connect piping to the ports of the suction manifold and connect the manifold to the fire pump inlet shall be corrosion-resistant metal.

3.11.5.3 Discharge section. The fire pump discharge section shall include a fire pump discharge manifold, fire pump discharge relief valve, discharge control valve, discharge outlets, piping, connections and discharge outlet drains as specified herein. All components and piping in the discharge section shall be austenitic or duplex stainless steel passivated in accordance with ASTM A 380,, or galvanized inside and out in accordance with ASTM A 525, G 90 coating. Brass, zinc coated metals, ot 400 series stainless steel shall not be used.

3.11.5.3.1 Fire pump discharge manifold. The fire pump discharge manifold shall be provided with the necessary discharge ports for attachment of valves and piping as specified herein. The ports shall be sized to accommodate the flow volumes required. The fire pump discharge manifold shall distribute discharge flows from the fire pump to the valves and piping.

3.11.5.3.2 Discharge relief valve. The discharge relief valve shall automatically control the discharge pressure according to the setting established by the pump operator. The design of the discharge relief valve shall provide that an increase of not more than 30-psig pump pressure shall occur when: (a) the pressure relief setting of the valve is 10 psig higher than the pump working pressure, and (b) all discharge valves are slowly closed. The relief valve shall be designed to operate at any fire pump working pressure above 75 psig when the fire pump is pumping at rated capacity. The design of the relief valve shall incorporate provisions for the attachment of an indicator light that will indicate discharge relief valve actuation. The location of the relief valve shall be such that remote adjustment of the discharge relief valve setting can be effected. Control of the relief valve and location of the indicator light shall be as specified in 3.11.6.3.1.8.

3.11.5.3.3 Discharge control valves. The discharge control valves shall be properly sized, balanced, corrosion-resistant, full flow valves designed for remote control. Controls for each discharge control valve shall be as specified in 3.11.6.3.1.6. Each control valve shall be labeled "discharge" and systematically numbered according to the corresponding discharge outlet and valve control. Numbers shall be affixed by corrosion-resistant, permanently attached tags.

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3.11.5.3.4 Discharge outlets. Discharge outlets that have NH thds or adapters that connect to hoses shall have machined ends to provide a seat for hose coupling gaskets or cap gaskets. Discharge outlets shall be designed into the truck as follows:

a. Four 2.5-inch NH discharge outlets; two on the streetside and two on the curbside of the truck. Each outlet shall be equipped with a 2.5-inch, 30- or 45-degree chrome plated elbow. Each elbow shall be equipped with a 2.5-to 1.5-inch NH chrome plated adapter. The adapter shall be capped with a 1.5-inch chrome plated cap which has an attaching chain.
b. One discharge outlet piped into the water tank and sized to flow not less than 50 gpm.
c. Two discharge outlets sized to flow not less than 250 gpm with an internal pressure drop of not more than 35 psig, at the small-hose bin (see 3.11.6.1.1) located above the fire pump. Each outlet shall be furnished with a swivel and a 1.5-inch NH fitting. The outlet shall be located to allow playout of the hose to either side of the truck. The outlet and swivel shall be located to allow easy connection of hose fitting.
d. One discharge outlet not less than 3-inches dia piped and valved to the aerial ladder/water tower.

The discharge outlets shall be systematically numbered according to the corresponding discharge valve and control. Numbers shall be affixed by austenitic or duplex stainless steel. Brass, zinc coated metals, or 400 series stainless steel shall not be used, permanently-attached labels adjacent to the outlet.

3.11.5.3.5 Discharge outlet piping. Each discharge outlet shall be piped to its applicable discharge control valve, using corrosion-resistant metal connections. The method of piping shall accommodate the discharge outlet drains specified in 3.11.5.3.6, as applicable.

3.11.5.3.6 Discharge outlet drains. The discharge outlets specified in 3.11.5.3.4a shall have a drain valve of not less than 0.75-inch dia installed at the lowest point in its piping to permit draining while the discharge control valve is closed. Drain valves shall be designed for remote control. Drain pipes from the drain valves shall be concealed behind truck body panels and shall discharge to a point below the bottom of the frame. Controls for the drain valves shall be as specified in 3.11.6.3.1.9.

3.11.6 Truck body. The truck body shall be chassis-mounted. The body shall provide for wheelhousings and running boards. The body shall also provide for the attachment of stowage units (hosebins, cabinets, etc.). The truck body shall be reinforced at all points of attachment. Rear wheel housings shall be designed to allow the use of snow chains.

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3.11.6.1 Integral stowage facilities. Integral stowage facilities shall be designed into the truck body. The integral stowage facilities shall consist of small-hose bins and large-hose bins. Design of the integral stowage facilities shall be as specified in 3.11.6.1.1 through 3.11.6.1.2. The integral stowage facilities shall be labeled to indicate stowed contents in accordance with the labeling requirements of MIL-STD-1472.

3.11.6.1.1 Small-hose bin. A small-hose stowage bin shall be designed to be located across the top of the truck body, above the fire pump, the suction and discharge manifolds, and the control valving. The small-hose bin shall be as long as the full width of the truck body. The small-hose bin shall have a removable metal divider that provides two equal small-hose stowage compartments. The size of the bin shall be large enough so that each compartment shall be capable of holding not less than 250 ft of 1.75-inch dia preconnected hose when the divider is in place. The bottom of the entire bin shall be removable when the divider is in place. The bottom of the entire bin shall be removable when the divider is removed. The opening created by removal of the bin bottom shall be sufficient to accommodate fire pump removal from the truck when the fire pump is disconnected from the suction and discharge sections. The removable hosebin bottom shall be designed as specified in 3.11.6.1.1.1. The metal bin bottom shall be protected by a hosebin floor as specified in 3.11.6.1.1.2. The interior of the bin shall be free of projecting nuts, sharp angles, or reinforcing brackets. Exposed bolt or rivet heads in the bin interior shall be of the oval type. Hose rollers shall be installed on the side and bottom openings of the bin to serve as guides when playing out hose and to prevent hose chafe. Hose rollers shall be austenitic or duplex stainless steel. Brass, zonc coated metals, or 400 series stainless steel shall not be used that has been passivated in accordance with ASTM A 380. Hose shall not be furnished.

3.11.6.1.1.1 Hosebin bottoms. Hosebin bottoms shall be full metal panels with crossmembers for support of a hosebin floor. The crossmembers shall have provisions for the attachment of hosebin floors. Bin bottoms shall have drain holes between crossmembers to prevent water retention in the bin. Bin bottoms shall be capable of supporting 200 lbs of deadweight on any 1 square ft of floor area without deformation.

3.11.6.1.1.2 Hosebin floors. Hosebin floors shall be slatted units. The slats shall be constructed of aluminum, or plastic-base, space-age, state-of-the art material. The slats shall be not less than 3.5 inches wide and have a finished thickness of 0.75 inch. The spacing between slats shall not be less than 0.625 inch nor more than 0.75 inch. The slats shall be fastened to the hosebin bottom crossmembers by countersunk screws or bolts. The slats of the floor shall be designed to permit easy removal of the bin bottom.

3.11.6.1.2 Large-hose bin. A large-hose bin shall be the inner top of the truck body, aft of the small-hose bin specified in 3.11.6.1.1. The large-hose bin shall not be wider than the distance between the rear wheelhousings and shall extend lengthwise of the body. The large-hose bin shall have a removable metal divider that separates the bin into two equal, lengthwise compartments. The size of the bin shall be large enough so that each compartment shall be capable of holding not less than 600 ft of 3-inch, double-jacketed, rubber-lined fire-hose. The bin bottom shall be as specified in 3.11.6.1.1.1 and the bin floor shall be

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as specified in 3.11.6.1.1.2. The water tank fill opening or cover and the water tank overflow drain and vent shall not extend into the large-hose bin. The interior of the bin shall be free from projecting nuts, sharp angles, or reinforcing brackets. Exposed bolt or rivet heads in the bin interior shall be of the oval type. Hose shall not be furnished.

3.11.6.2 Attached equipment cabinets. Equipment cabinets shall be attached to the truck body as specified in 3.11.6.2.1 and 3.11.6.2.2. Five cabinets shall be located on the streetside of the truck and five cabinets shall be located on the curbside of the truck. The streetside cabinets shall be arranged as two lower and three upper cabinets. Cabinet construction shall be as specified in 3.11.6.2.3.

3.11.6.2.1 Streetside cabinets. The two lower streetside cabinets shall have interior compartments that are not less than 34 inches high from floor to top and not less than 14 inches deep from door to back. The individual lower cabinet widths shall be the maximum width practicable that does not interfere with the pump operator's panel (see 3.11.6.3), the wheelhousing, the fuel inlet, or the outriggers (see 3.11.11.6), but not less than 54 interior inches. Filler panels and skirting shall be installed between lower cabinets to provide a flush side and top. The four upper streetside cabinets shall be vertically flush with the lower cabinet doors and skirting panels. The backs of the upper cabinets shall be flush with, and attach to, the exterior of the large-hose bin. The height of the upper cabinets shall extend from the top of the lower cabinets and filler panel to the top edge of the large-hose bin. The three upper cabinets shall have a collective width that is not less than the end-to-end distance of the two lower cabinets. The tops of the upper cabinets shall be continuous, without gaps, and shall conform to 3.9.3.

3.11.6.2.2 Curbside cabinets. The curbside cabinets shall be identical to, but obverse copies of the streetside cabinets. The curbside cabinet tops shall be connected with a filler panel to provide a continuous top, without any gaps, which shall conform to 3.9.3. Flush skirting shall be provided as on the streetside. One cabinet shall be equipped with brackets to securely hold not less than eight 30 minute air bottles. The bottles shall not be furnished.

3.11.6.2.3 Cabinet construction. The cabinets shall be constructed as specified in 3.11.6.2.3.1 through 3.11.6.2.3.5.

3.11.6.2.3.1 Cabinet bottoms and floors. All cabinets shall have bottoms and floors that are constructed as specified for hosebins in 3.11.6.1.1.1 and 3.11.6.1.1.2.

3.11.6.2.3.2 Cabinet doors. Cabinet doors shall be of the flush, box-type and double gasketed. Door handles shall be recessed "D"-ring-type, conforming to MIL-STD-1472 requirements for handles and grasp area, with two-point latching. Door hinges shall be full length, piano-type, of corrosion-resistant metal. Cabinets with door openings that exceed 30 inches in width shall be provided with double doors, but only the door hinged on the edge toward the front of the truck is required to have a handle. Cabinet doors without handles shall have two-point latching, with the latch release mechanism on the interior of the door. All cabinet doors with exterior handles shall be hinged on the edge toward the front

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of the truck. All doors shall be equipped with spring-loaded, hold-open devices. Drip rails shall be provided over all doors. Cabinet doors shall have label pockets on their interior where an inventory of cabinet contents can be stowed. All cabinet doors with exterior handles shall meet the KKK specifications for patient carrying components.

3.11.6.2.3.3 Cabinet lighting. The cabinets shall be automatically illuminated when its door is opened. A warning light, mounted on the dashboard in the driver's compartment, shall be wired through the automatic door switch to signal the driver that a cabinet door is ajar.

3.11.6.2.3.4 Cabinet ventilation and drainage. All cabinets shall be provided with a means of ventilation that will not admit water from rain, road splash, or firefighting operations. All cabinet seams shall be sealed against the entry of water. The top surfaces and filler panels of the lower cabinets shall be constructed to prevent water accumulation and treated to prevent corrosion at the cabinet and panel interface. Spacers shall be incorporated into the lower streetside cabinet mountings to assure water drainage and air circulation behind cabinets. One way drain holes shall be provided.

3.11.6.2.3.5 Cabinet mountings. Cabinets shall be constructed with reinforcement at all mounting points and the method of attachment shall be adequate to support the weight of not less than three 175-lb fire fighters on their top surfaces without evidence of fracture, warp, or sag at the mounting points.

3.11.6.3 Pump operator's panel. A pump operator's panel shall be provided on the streetside of the truck body. The panel shall be located forward of the streetside cabinets. The pump operator's panel shall accommodate the streetside inlets and outlets specified herein, and the controls, gages, and instruments specified in 3.11.6.3.1 through 3.11.6.3.3.3. The inlets, outlets, controls, gages, and instruments on the pump operator's panel shall be labelled in accordance with the labeling requirements of MIL-STD-1472. The controls shall be in accordance with the requirements for controls of MIL-STD-1472. Controls shall be located no higher than 73 inches and no lower than 36 inches above the ground. The pump panel shall be polished stainless steel for the street and curb side of the truck.

3.11.6.3.1 Pump operator's controls. The controls mounted on the pump operator's panel shall have plates or collars installed at the point where they emerge through the panel. The controls shall be as specified in 3.11.6.3.1.1 through 3.11.6.3.1.9.

3.11.6.3.1.1 Engine throttle control. The engine throttle control on the pump operator's panel shall be of the screw-locking, center-pushbutton-release type. The throttle control body shall hold the control lever in any desired position and shall not be affected by vibration or the pull of the accelerator spring. The engine throttle shall control engine rpm from manufacturer's recommended idle to maximum.

3.11.6.3.1.2 Heat exchanger control. A rotary or linear "on-off" control for the heat exchanger shall be mounted on the pump operator's panel.

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3.11.6.3.1.3 Foam-liquid control. The foam-liquid control on the pump operator's panel shall be a rotary control. A dial-and-pointer arrangement capable of calibration shall indicate regulation of discharge percentage or proportioning of foam liquid to total solution flow. An instruction plate with chart indicating the proper setting shall be located on the pump operator's panel adjacent to the control.

3.11.6.3.1.4 Water tank suction controls. The water tank shutoff valve control on the pump operator's panel shall be a linear control. The drain valve control for draining the water tank suction piping shall be located on the pump operator's panel adjacent to the water tank shutoff valve control.

3.11.6.3.1.5 Small suction inlet controls. The controls for the small suction inlets shall be mounted on the pump operator's panel and shall be an integral part of the suction valve. The controls shall be linear with twist-locking handles in all positions.

3.11.6.3.1.6 Discharge valve controls. The controls for the discharge valves shall be linear (push-pull). All of the discharge valve controls shall be symmetrically grouped. The controls shall be numbered to correspond to the applicable discharge outlet and control valve. The linear controls shall conform to Akron style 1488 or equal.

3.11.6.3.1.7 Fire pump drive control. A control for the fire pump drive may be mounted on the pump operator's panel. If a fire pump drive control is not mounted on the pump operator's panel, it shall be mounted on or under the dashboard in the driver's compartment of the cab. The pump drive control shall be positive locking in either position. Mounting in either position shall incorporate an indicator light to indicate pump engagement.

3.11.6.3.1.8 Discharge relief valve control. The control for the discharge relief valve on the pump operator's panel shall be a rotary control. An indicator light shall be connected to the discharge relief valve and mounted on the pump operator panel adjacent to the discharge relief valve control.

3.11.6.3.1.9 Discharge outlet drain controls. The controls for the discharge outlet drains shall be mounted on the pump operator's panel beneath the associated discharge valve control.
Each valve shall be labeled to indicate the associated discharge valve.

3.11.6.3.2 Pump operator's gages. The gages mounted on the pump operator's panel shall have plates or collars installed where they emerge through the panel. The gages shall be as specified in 3.11.6.3.2.1 through 3.11.6.3.2.5.

3.11.6.3.2.1 Master pressure gage. A compound master pressure gage on the pump operator's panel shall have a dial pressure range of 30 to not less than 600 psig. The dial face of the gage shall not be less than 3.5-inches dia. The gage shall be provided with a device to eliminate indications of pressure fluctuation (glycerin-filled gage is acceptable).

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3.11.6.3.2.2 Master suction gage. A compound master suction gage on the pump operator's panel shall have a dial pressure range of 30 to not less than 600 psig. The dial face of the gage shall not be less than 3.5-inches dia. The gage shall be provided with a device to eliminate indications of pressure fluctuations (glycerin-filled gage is acceptable). The master pressure gage and master suction gage shall be mounted next to each other.

3.11.6.3.2.3 Discharge outlet gages. A separate pressure gage shall be provided for each discharge outlet. Each gage shall be provided with a device to eliminate indications of pressure fluctuation (glycerin-filled gages are acceptable). The gages shall not be less than 2.5-inches dia. The gages shall be grouped in accordance with MIL-STD-1472 control requirements above the controls associated with the discharge outlets.

3.11.6.3.2.4 Engine and transmission operating gages. Engine and transmission operating gages shall be provided on the pump operator's panel, located to the left of the pump control panel, and separated from it by not less than 2 inches, as follows:

a. Engine oil pressure
b. Engine coolant temperature
c. Transmission oil temperature
d. Fuel.

The gages shall not be less than 2.5-inches dia. The gages shall be symmetrically grouped in proximity and above the engine throttle control and pump drive control in accordance with MIL-STD-1472 visual displays and control groupings. Each gage shall be provided with a remote indicating light to monitor conditions as specified in 3.11.2.1.2.3a, b, and c.

3.11.6.3.2.5 Water tank gage. A water tank gage that indicates the water level in the water tank shall be mounted on the pump operator's panel.

3.11.6.3.3 Pump operator's instruments. The instruments mounted on the pump operator's panel shall have plates or collars installed where they emerge through the panel. The instruments shall include an engine tachometer, voltmeter, and two test connections.

3.11.6.3.3.1 Engine tachometer. An engine tachometer shall be provided on the pump operator's panel. The tachometer shall be identical to and interchangeable with the tachometer specified in 3.11.2.1.2.1b. The tachometer shall be located directly above the engine throttle control. The tachometer shall be equipped with the means for checking the tachometer accuracy with a hand-held revolution counter without disconnecting the tachometer.

3.11.6.3.3.2 Voltmeter. A voltmeter shall be provided on the pump operator's panel. The voltmeter shall not be less than nominal 2.5-inches dia. The voltmeter shall be located in proximity to the engine operating gages specified in 3.11.6.3.2.4.

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3.11.6.3.3.3 Test connections. Two 0.25-inch NPT female test connections for test gage attachment shall be provided on the pump operator's panel. The connections shall be located near the master suction and master pressure gages. The connections shall be piped to the respective suction and discharge manifolds and labelled accordingly. The connections shall be plugged with removable NPT male plugs of chrome plated material.

3.11.6.4 Pullout treadplate. An aluminum pullout treadplate shall be provided beneath the running board at the pump operator's panel. The treadplate shall be capable of supporting not less than 400 lbs on its outermost edge. The width of the treadplate shall be not less than 80 percent of the width of the pump operator's panel. The front-to-back depth of the treadplate shall not be less than the running board width. A positive-locking mechanism shall be provided to secure the treadplate in the closed (retracted) position.

3.11.7 Electrical system. The electrical system for the truck shall be wired as specified in 3.5.3. Wiring for the 110 Vac system shall be run separately from the 12-vdc wiring. If manuals are furnished with the fire truck, a complete wiring diagram of the entire electrical system shall be incorporated in any operating or maintenance manual furnished with the truck.

3.11.7.1 Electrical equipment. All electrical equipment of the truck shall be 12 vdc, except for the 110 Vac receptacles, the wiring between them, and the 300-W floodlights specified in 3.11.7.4.4. All electrical equipment shall meet the criteria set forth in MIL-STD-454, requirement 1. All exposed parts which are electrically energized shall be insulated, fully enclosed, or guarded. Components shall be located so as not to be adversely affected by engine heat. All electrical components shall be water-resistant and designed so that their normal function will not be impaired by entrance of water during fire-fighting operations, heavy rains, road splash, condensation, or from water spray resulting from other causes. All noncurrent-carrying metal parts shall be grounded. Any 6-volt (v) equipment, paired in series with resistors or tapped to the midpoint of one of the batteries, shall not be used in the truck.

3.11.7.2 Storage batteries. Not less than two 12-v storage batteries shall be furnished with each truck. The batteries shall be stowed in a battery tray, or trays, designed of sufficient size to hold and store the batteries specified. Batteries shall be shipped with the truck in a charged condition.

3.11.7.2.1 Battery tray(s). The battery tray(s) may be either drawout or fixed type. If a drawout type is provided on the truck, there shall be a means of positive locking of the tray(s) in the retracted position to prevent battery cable chafing. If a fixed type is provided on the truck, mounting on the streetside running board shall be permitted, provided no part of the tray(s) extends beyond the attached equipment cabinets. Batteries shall be securely held in place by means of a removable locking device and shall be protected against engine and exhaust heat. The battery tray(s) shall be ventilated and treated against acid fumes and spillage. The battery terminals shall be provided with guards to prevent accidental shorting. A pictorial diagram of battery connections shall be provided which is readable when batteries are being serviced.

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3.11.7.3 Battery circuits. Battery circuits shall conform to 3.11.7.3.1, 3.11.7.3.2 and the wiring diagram titled "Suggested Application of Dual Batteries to Single Ignition Engine" in the appendix to NFPA Code No. 1901.

3.11.7.3.1 Battery charging circuits, station. Each battery shall be connected to an electrical receptacle for station charging. The receptacles shall be mounted on the truck in such a location that forward movement of the truck will result in an automatic disconnect without damage to the receptacles aboard the truck or the mating plug and its attached cables from the station-charging unit. The mating plug shall be furnished with the truck.

3.11.7.3.2 Battery charging circuits, truck. The battery circuits between the generating system (see 3.11.1.7) and the batteries shall have reverse polarity devices that protect the alternator/voltage regulator from damage when the batteries are connected in reverse polarity. The battery circuits shall have a control system installed whereby:

a. Either battery may be used for:

   (1) Ignition and lights alone.
   (2) Starting alone.
   (3) Starting, ignition, and lights.

b. Both batteries may be used simultaneously in engine starting.
c. Neither lighting wiring nor ignition wiring will be connected to both batteries except when both starter switches are operated.
d. The alternator/voltage regulator shall be connected to supply the battery being used for ignition and lights.

3.11.7.4 Lighting system. The truck lighting system shall be designed to include truck operating lights, equipment operating lights, warning lights, and the 110-vac floodlights.

3.11.7.4.1 Truck operating lights. All truck operating lights, reflectors, and wiring shall conform to the DOT Motor Carrier Safety Regulations, except as otherwise specified herein:

a. DOT identification lamps are required.
b. Clearance and side-marker lamps shall be combined and mounted to indicate width of the truck.
c. Rear-mounted lamps shall be located so that they shall not be obscured by equipment normally carried on the platform.
d. Front turn-signal units shall be single-faced, flush-mounted. The turn-signal controller shall be the self-cancelling type and shall be mounted on the steering column; shall operate visual indicators on the dash; and shall be wired through the ignition system.
e. Stoplights, tail lights, rear turn-signal and backup lights shall be flush-mounted on the rear of the body. Backup lights shall not be less than 5-inches dia and shall automatically be activated when the transmission is shifted into reverse.

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3.11.7.4.2 Equipment operating lights. The truck equipment operating lights shall be as follows:

a. Two automotive-type sealed-beam spotlights shall be located on each side of the windshield, with controls extended inside the cab for "on-off" control and movement of the lights through horizontal and vertical angles.
b. Pump operator's panel lights shall be provided and enclosed within hoods or covers to completely illuminate the panel to not less than 30 foot-candles and to prevent glare.
c. Two sealed-beam hose pickup lights shall be mounted on the rear of the truck. Hose pickup lights shall not be mounted on or attached to any handrail or grabrail.
d. Step-lights shall be provided for the cab and jump-seat steps and controlled by the headlight switch. Step- lights shall be hooded or shielded for downward illumination only.
e. Two engine-compartment lights shall be provided to illuminate both sides of the engine. Both lights shall be shielded and controlled by one switch on the dash.

3.11.7.4.3 Warning lights. Warning lights shall conform to NFPA Code No. 1901. Two motorized, oscillating, sine-wave-pattern warning lights shall be mounted beneath the windshield in equally spaced positions on the front of the cab. Two strobe-type warning lights shall be mounted on the rear of the truck, outside the hose pickup lights. Two warning lights containing not less than three sealed-beam rotating lamps shall be centric mounted on top of the cab on either side of the aerial ladder/water tower riding position. All warning lights shall have red lenses.

3.11.7.4.4 Floodlights. Two portable, 300-W, 110-vac floodlights, with three wire-grounded connectors (circle D, portable light No. 150 and model P-23 holddown clamps or equal), shall be mounted on either side of the truck above the small-hose bins. The floodlights shall have a pigtail wire not less than 4 feet long for connection of the light to the electrical outlet.

3.11.7.4.5 Floodlight receptacles. There shall be five 110- Vac receptacles mounted on truck body panels. There shall be one receptacle mounted on each side of the truck, adjacent to the fire pump, and two receptacles mounted near the top of the truck body, above the fire pump for connection of the lights specified in 3.11.7.4.4. A receptacle shall be provided at the tip of the ladder with a control switch mounted in the ladder control station. The receptacles shall be dual, grounded, and weatherproof, with spring-loaded covers. Wiring between the generator and the receptacles shall be of sufficient gage, fused, properly supported, and protected from injury. Each receptacle shall be labeled 115 vac, 60 Hz, and shall conform to the labeling requirements of MIL-STD-1472.

3.11.7.5 Lighting system controls. Lighting system controls not otherwise specified herein, shall be as specified in 3.11.7.5.1 through 3.11.7.5.3. Labels and marking for lighting system controls shall conform to the labeling requirements of MIL-STD-1472.

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3.11.7.5.1 Headlight control. The headlight control shall be a push-pull type switch that shall activate headlights, tail lights, and step-lights. Rotation of the headlight switch shall dim or brighten the gage and instrument lights specified in 3.11.2.1.2.1, and shall activate the dome light in the driver's compartment at the end of the rotary dimming revolution. The headlight switch shall be mounted on the dash in the driver's compartment.

3.11.7.5.2 Warning-light controls. Warning-light switches, that will activate the warning lights specified in 3.11.7.4.3 shall be single-pole, single-throw, rocker, circuit-breaker type switches, with indicating light for the "on position. The switches shall be mounted on the dash in the driver's compartment within the driver's functional hand reach in accordance with SAE J898, in a group of three adjacent, identical switches, independent of other electrical equipment controls. Shielding of the indicating lights shall be provided if necessary to prevent glare on the windshield.

3.11.7.5.3 Floodlight receptacle controls. The two upper floodlight receptacles shall be controlled by two "on-off" switches. The switches shall be mounted on the dash in the driver's compartment.

3.11.8 Audible warning system. The design of the truck shall incorporate the audible warning system specified in 3.11.8.1 through 3.11.8.4.

3.11.8.1 Sirens. The truck shall be equipped with an European, 3 mode type electronic siren and public address combination. The electronic siren shall be furnished with a microphone and all wiring applicable for connection to a two-way radio. A single, 6-inch-dia chromium plated speaker and bracket shall be mounted between the sine-wave warning lights specified in 3.11.7.4.3. A Federal Sign and Signal Corporation siren model Q2B, or equal, shall be mounted on the left side of the truck on a front bumper extension plate. Controls to activate the siren shall be two foot-operated buttons designed in conformance with MIL-STD-1472 requirements for foot- operated switches. One button shall be located for driver left foot and one for passenger right foot use. An electric brake control for the siren shall be mounted on the dash in a location equally accessible to driver or passenger. The electric siren brake shall be designed so that it cannot brake a siren that is turned on. A 12-inch bronze, chrome plated, hand-operated bell shall be mounted on the right side of the truck on a front bumper extension plate.

3.11.8.2 Horns. Two tone-adjustable, trumpet-type, air-activated horns shall be mounted on the cab roof outboard of the warning lights specified in 3.11.7.4.3. The horns shall be powered with air from the brake air tank. The horns shall be controlled by a pull chain that is equally accessible to driver or passenger.

3.11.8.3 Buzzer. A buzzer shall be installed in the driver's compartment. The buzzer shall be controlled by a waterproof, pushbutton switch mounted on the rear of the truck, above the back steps, in a position that permits personnel to signal the driver while standing on the ground. The buzzer shall be audible when the truck is operated in the noisiest manner.

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3.11.8.4 Backup warning. An electrically operated backup warning, in accordance with SAE J994, shall be installed to automatically operate when the truck transmission is shifted into reverse.

3.11.9 Handrails and grabrails. Handrails and grabrails shall be incorporated into the truck. The handrails and grabrails shall be 1.25-inch dia, chrome plated or stainless steel conforming with MIL-STD-1472 handle and grasp area requirements for gloved hands, and be made of heavy tubing mounted on brackets. The tubing shall be secured in the bracket to prevent rotation. As a minimum, the following handrails and grabrails shall be installed aboard the truck to assist personnel climbing aboard the truck and getting to the turn table:

a. Two vertical grabrails, on either side of the truck body at the rear step, to assist personnel climbing to the turn table.
b. Two handrails, located atop the large-hose bins, on the left and right sides of the truck. These handrails shall be the same length as the bins to which they are attached.
c. One horizontal grabrail, not less than 12 inches long shall be mounted under each front windshield to assist personnel in getting up under the front bumper plate extension to clean the windshields.
d. Two vertical grabrails shall be located on either side of the jump-seat area to assist jump-seat passengers up the steps into the jump-seat area.
e. Two vertical grabrails shall be located on either side of the driver's compartment to assist driver and passenger entering the compartment.
f. Two horizontal handrails on the interior of each of the cab doors.
g. Handrails around the perimeter of the turntable (see 3.11.11.2) not less than 30 inches above the deck of the turntable, with an opening for personnel access to the turntable and aerial ladder/water tower combination.
h. Two grabrails, one under each front window, shall be provided to aid personnel in getting onto the front bumper extension plate for purposes of cleaning the windshield.

3.11.10 Stowed components. The stowed components specified in 3.11.10.1 through 3.11.10.8 shall be furnished with the truck. The holders, brackets, or mounting pads used to mount the stowed components on the truck shall be positive-locking, quick-release devices that keep the stowed components secure under all running conditions, yet permit quick removal.

3.11.10.1 Soft suction hose. One 12 ft length of 6-inch-dia, commercial-grade, soft suction hose shall be stowed in the soft suction hose compartment specified in 3.11.2.3. The hose shall be equipped with a 4.5-inch, long-handled, swivel-type, female-gasketed coupling on one end and a 6-inch, long-handled, swivel-type, female-gasketed coupling on the other. The hose shall be connected to the front suction inlet. The hose coupling when connected shall not extend past the bumper.

3.11.10.2 Reducer. One 6-inch NH to 4.5-inch NH reducer, with double-female, double-swivel, long-handled, gasketed couplings shall be mounted on the inside of a streetside cabinet's door.

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3.11.10.3 Ladders. Ladders shall be in accordance with NFPA Code No. 1931. Access to the ladders and hard suction hose shall be from the rear of the truck. The access opening shall be provided with a roll up door. The ladders furnished shall be as follows:

a. One 24-ft, two-section, extension type.
b. One 14-ft roof-type.
c. One 10-ft attic-type.

3.11.10.3.1 Ladder stowage. The ladders shall be stowed between the frame rails below the turntable. The ladders shall be removed from the rear of the truck. The ladders shall be stowed and held in place to prevent chafe and rattle by an all metal single control adjustable type device.

3.11.10.4 Axes. Two axes shall be stowed in the jump-seat area on the exterior of the vertical cab extension. The axe mounted on the curbside shall be a pick-head fire axe with a fiberglass handle. The axe mounted on the streetside shall be a flat-head, Dayton-pattern fire axe with a fiberglass handle. A pick-head fire axe shall be stowed on the fly section end of the aerial ladder/water tower. Both pick-head axes shall be provided with a cover for the pick end. The cover will stay with the mounting device when the axe is removed.

3.11.10.5 Pike pole. One straight, plain-point, and hook pike pole with a 6-ft fiberglass handle shall be stowed on the aerial ladder/water tower at the fly section end. Two 8 foot and one 12 foot fiberglass handles with plain point and hookpike poles shall be stowed with the ground ladders.

3.11.10.6 Wheel chocks. Four wheel chocks shall be furnished and stowed in brackets beneath the running boards at the jump-seat steps.

3.11.10.7 Hard suction hose. When specified (see 6.2), one 10-ft length of 6-inch-dia, commercial-grade, hard suction hose shall be stowed in each suction hose tray or tube (see 3.11.10.7.1). The hose shall have 6-inch-dia NH, long-handled, female couplings on one end and 6-inch-dia NH, rocker-pin-lug, male couplings on the other.

3.11.10.7.1 Suction hose trays or tubes. When hard suction hose is required (see 6.2), the truck shall be equipped with two suction hose trays or tubes. The suction hose trays or tubes shall be sized to stow 10 ft lengths of 6-inch-dia hard suction hose with couplings. The trays or tubes shall be installed within the truck body. The fasteners for the trays or tubes shall facilitate their removal for access to the water tank top. The trays or tubes shall be equipped with positive-locking mechanical devices to prevent hose movement under all road conditions. The trays or tubes shall be equipped with drain holes over their entire length. The finish of the trays or tubes shall be such as to prevent hose chafe. The installation fasteners, hose-locking devices, and the trays or tubes shall be of corrosion-resistant steel that has been passivated in accordance with QQ-P-35. Access to the trays or tubes shall be provided with doors as specified in 3.11.6.2.3.2, except that door locks shall not be furnished.

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3.11.10.8 Strainer. When hard suction hose is furnished, one strainer, with a 6-inch-dia NH female coupling, shall be stowed on the step adjacent to the pump operator's panel on the curb side of the truck.

3.11.11 Aerial ladder/water tower. The truck shall be equipped with an aft-mounted, hydraulic-powered, telescoping, aerial ladder. The aerial ladder shall be a three- or four-section unit that extends not less than 100 ft when measured as specified in NFPA Code No. 1901. The aerial ladder shall be equipped with a remote nozzle (see 3.11.11.5) and other equipment specified herein to form an aerial ladder/water tower. The aerial ladder/water tower shall be capable of full extension, elevation, and rotation with a 400-lb deadweight load attached to the tip of the fly-section end and shall resist and support the reaction of a remote nozzle flowing not less than 1000 gpm. The aerial ladder/water tower shall be capable of being depressed to not less than 2 degrees below and at least 75 degrees above the horizontal while discharging water at rated capacity.

3.11.11.1 Hydraulic system. The hydraulic system used to actuate the aerial ladder/water tower shall consist of all fluidpower components as specified herein. The system fluid shall be oil conforming to MIL-L-2104, grade 10, or MIL-L-46167. All hydraulic system component materials shall be compatible with these oils. The hydraulic system shall have an oil capacity and a working pressure that is sufficient to operate the water tower as specified herein. The working pressure of the hydraulic system shall not exceed 3000 psi. System pressure shall be provided by an engine-driven pump that is powered by the truck engine. An interconnecting device between the hydraulic system and the truck engine throttle shall be set by the manufacturer to provide the optimum performance of the ladder operation. The hydraulic system shall incorporate a 12 Vdc electric motor driven backup pump to provide a pressure source under emergency conditions.

3.11.11.1.1 Hydraulic fittings. The hydraulic fittings on pumps, valves, and cylinders shall be either O-ring, straight- thread type, as specified in SAE J514, or the 4-bolt flange type, as specified in SAE J518; however, only 4-bolt flange type fittings specified in SAE J518 shall be used for ports larger than 1 inch.

3.11.11.1.2 Hydraulic lines. Hydraulic lines shall consist of tubing, flexible hose, and steel fittings. Hose or combinations of hose and tubing shall be used between components that are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide a minimum number of bends, ease of maintenance, maximum protection, and to avoid chafing. The hose installation, including angle of fittings and location, shall cause no concentrated stress on the hose at the fitting; also, hose shall be supported as necessary to eliminate sagging at the fittings. All lines except suction lines or cylinder bleeder lines shall be considered pressure lines.

3.11.11.1.2.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 or MIL-H-52471. Permanently attached pressure hose couplings are acceptable on original equipment hose assemblies. The pressure that is equal to

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or greater than the hydraulic circuit maximum relief-valve setting. Hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.11.11.1.2.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525. The pressure tube sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. Tubing shall have a burst pressure that is not less than three times the system maximum working pressure. All tubing shall be securely clamped and all bends shall be smooth without flattening (see 6.4.4), kinking, or wrinkling of the tube.

3.11.11.1.2.3 Pressure tube fittings. Pressure tube fittings shall conform to SAE J514. Jump-size tees may be used. Pressure tube fitting sizes shall be limited to the sizes specified for pressure tubing. Pressure tube fittings shall have a burst pressure that is not less than three times the system maximum working pressure.

3.11.11.1.2.4 Suction lines. All pump inlet suction lines shall incorporate a flexible means of absorbing vibration or movement between the pump and reservoir. The suction line shall be sized so the pump inlet pressure does not drop below manufacturer's rating at 140 °F ±10 °F oil temperature and at controlled engine speed. The suction hose shall conform to SAE J517, 100R4. Suction hose clamps shall be of the double-wrap, stainless-steel type in accordance with SAE J536, type F, style 4.

3.11.11.1.2.5 Return lines. Return lines shall be in accordance with SAE J517.

3.11.11.1.3 Hydraulic pump. The pump manufacturer's 1000-hour pump contaminant tolerance profile for the pump(s) shall not fall below or cross the B_{10} curve of the hydraulic system filters, when the filters have been tested in accordance with MIL-F-8815/23; or the pump Omega rating shall be less than the filter Beta 10 rating.

3.11.11.1.4 Hydraulic system filtration. Hydraulic system filters shall be furnished in the pressure line, return line, or both.

3.11.11.1.4.1 Return-line filters. The filter element shall maintain a full-flow condition for the duration of all testing specified herein. The filter bypass may open for system oil at a system temperature that is less than 70 °F. All return oil shall pass through the filter, except relief oil, which may return directly to the reservoir. The filter shall be located so that the element may be removed for examination and replacement without spilling or draining oil from the reservoir. At maximum system oil flow through the filter, including ladder/tower depression at maximum controlled engine speed and an oil temperature of 150 °F ±5 °F, the pressure drop across a new filter assembly shall not exceed 10 psi. An automatic bypass shall be provided in the system filter and shall open at not less than 10 psig more than the pressure drop measured across the new filter assembly, and shall permit full flow at a pressure drop of not

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more than 35 psig. The filter housing shall have a proof pressure of not less than 100 psig without leakage, damage, or permanent deformation. The filter assembly shall be equipped with provisions for a remotely mounted contamination indicator that will indicate when the filter is clean and when the filter is bypassing.

3.11.11.1.4.2 Pressure line filter. The pressure line filter shall be as specified in 3.11.11.1.4.1, except that the proof pressure of the filter housing shall not be less than 150 percent of the maximum system operating pressure.

3.11.11.1.5 Hydraulic fluid reservoir. A hydraulic reservoir shall be furnished to contain hydraulic system oil. The reservoir shall be mounted to resist vibrations that produce leaks, loosening of fittings or fasteners, brinelling of material, or cracks. The reservoir shall have sufficient capacity to prevent air from entering the hydraulic lines when all pistons are fully extended and sufficient free air capacity to prevent oil from being discharged through the reservoir air vent when maximum return oil is being surged into the reservoir. A sightglass or dipstick, with appropriate "Add" and "Full" marks, shall indicate the correct oil levels when the truck is standing on level ground and the water tower is in the riding position. A filler opening shall be located to provide for filling of the reservoir from a 5-gal container conforming to MIL-C-1283 while standing on the ground or a riding surface of the truck. The neck of the filler opening shall contain a removable 50-mesh-or-finer screen strainer. The filler tube inside dia shall be not less than 1 inch. The filler opening shall be sealed with a removable filler cap attached to the truck with a chain. The suction line connections shall be not less than 2 inches above the reservoir's bottom. Return-line connections shall discharge into the reservoir below oil level. The suction and return-line connections shall be located on different sides of the reservoir, and interior baffling inside the reservoir shall provide a tortuous flow path that will prevent a direct current between ports that connect the pump and ports that connect return lines. The reservoir shall be equipped with one or more access holes, with covers, for inspection and cleaning.

3.11.11.1.6 Pressure relief valve. A pressure relief valve shall be provided to limit pump pressure. Pressure equal to the relief-valve setting shall be generated only when a cylinder is bottomed. The relief-valve setting shall not exceed the pump manufacturer's rating.

3.11.11.1.7 Cylinders. All hydraulic cylinders that actuate implements shall be fabricated of seamless steel tube or welded and redrawn tube. All welding and brazing of the cylinder within the zone of piston operation shall be done before final machine honing or rolling of the cylinder. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall be equipped with an exclusion device that prevents foreign material and fluids from entering around the piston rod and damaging seals and other cylinder components. Each piston shall have a bearing surface concentric with the piston to carry piston sideloads against the cylinder wall. The bearing surface shall be of a material that will not scar, score, or gall the inside of the cylinder. Cylinder

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rods shall be hard chrome plated. All cylinders shall perform without buckling or bending and shall withstand a pressure that is equal to 1.5 times the relief-valve setting without evidence of rupture, permanent deformation, internal or external leakage, or any form of visible damage. Cylinders shall be located for ease of maintenance and replacement.

3.11.11.1.8 Hydraulic system cleanliness. The particle counts in the hydraulic system after installation of all fluidpower components, and flushing and cleaning of the system before delivery, shall not exceed 1000 particles per milliliter greater than 10 micrometers and 20 particles per milliliter greater than 20 micrometers.

3.11.11.1.9 Hydraulic leakage. After completion of the pumping stability test (see 4.5.4), the hydraulic system leakage shall not exceed SAE J1176 class 4 for dust-free conditions and class 4D for dusty conditions.

3.11.11.2 Turntable. The turntable of the aerial ladder/water tower shall be rotated by a hydraulic motor driving through a ring and pinion gear set. The turntable shall be capable of rotating in continuous 360-degree turns in either direction. The turntable top shall be not less than 60 inches in dia and shall have raised pivots for mounting the ladder that will permit depression below the horizontal. Control of the turntable rotation shall be incorporated into the one-hand-operated, ladder/tower hydraulic control (see 3.11.11.7.1) mounted on the control station (see 3.11.11.7). Steps to permit personnel access to the turntable from the platform shall be provided on the curb and street side at the rear of the truck with appropriate grabrails.

3.11.11.3 Aerial ladder/water tower construction. The aerial ladder/water tower combination shall be constructed of welded, clad form, high-strength steel or aluminum. Each section shall be trussed diagonally, vertically, and horizontally. There shall be a minimum 18-inch width between the ladder rails when measured at the narrowest point between the rails or the trussing. Ladder rungs shall be welded through a minimum of each inside section beam to provide torsional rigidity to the ladder sections. Ladder rungs shall be covered with heavy-duty, rubber sheaths. The sheaths shall not rotate on the ladder rungs. Rung spacing, center to center, shall be as specified in NFPA Code No. 1931. Handrails shall be provided for the full length of the ladder. Handrails shall extend a minimum of 18 inches above the ladder rungs on the base section. When the ladder is horizontal and fully retracted, the handrails of each section shall be in an identical horizontal plane. Two folding steps shall be placed near the top of the ladder for use by the remote nozzle (see 3.11.11.5) operator. If the top rung of the fly section end is too large to attach the hook of a pompier or lifebelt, a secure attachment device shall be provided for the operator to tie-in. Wear pads shall be provided between telescopic sections of the ladder.

3.11.11.4 Aerial ladder/water tower water system. Water for the ladder pipe shall be supplied by the fire pump. The piping shall be valved at the discharge manifold with an appropriate sized valve that is controlled at the pump operator's panel. The piping shall then be routed through a master control valve at the aerial ladder/water tower control station before attaching to the ladder pipe. The piping between the valve at the discharge manifold and the master

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control valve shall incorporate a drain, as specified in 3.11.5.3.6, and a drain control, as specified in 3.11.6.3.1.9. Ladder piping and connections between the master control valve and the remote control nozzle shall be capable of providing a discharge rate of 1000 gpm of water when:

a. The turntable is rotated through any horizontal arc up to 360 degrees.
b. The aerial ladder/water tower is either fully extended or fully retracted.
c. The boom and ladder is completely horizontal or at a vertical angle of not less than 75 degrees.
d. The truck is sitting on a level, hard surface and the outriggers (see 3.11.11.6) are in the operating position.

3.11.11.4.1 Master control valve. The master control valve shall be designed for remote or through-panel control. The valve shall be of corrosion-resistant metal. The control for the master control valve shall be mounted on the control station.

3.11.11.5 Remote control nozzle. A remote control nozzle (monitor) shall be mounted on the fly section end of the aerial ladder/water tower. The monitor shall be equipped with a remotely controlled nozzle tip (stream shaper) that will provide a uniform stream of water at any flowrate between 300 gpm and 1000 gpm. The monitor shall be capable of being elevated or depressed in a vertical plane of not less than 250 degrees. The monitor shall be capable of being swept laterally through a 180 degree arc (90 degrees to either side of center). The functions of elevation and depression in the vertical plane shall be completely independent of the sweep function in the horizontal plane. The functions of nozzle elevation, depression, and sweep shall be hydraulically actuated. A set of controls for the monitor shall be mounted near the monitor that can be operated from the top of the ladder and a second set of controls shall be mounted on the control station.

3.11.11.5.1 Remote nozzle protection. The fly section end of the aerial ladder/water tower shall be equipped with protective brackets on either side that prevent damage to the monitor and tip when it is fully depressed against the underside of the fly section end.

3.11.11.6 Outriggers. The truck shall be equipped with hydraulically actuated outriggers. The outriggers shall stabilize the truck when the aerial ladder/water tower is raised, supporting a 400-lb load at the top of the ladder and discharging water. The outriggers shall be self-locking and equipped with flashing red lights (at a two per second rate) to warn of outrigger extension. The red warning lights shall activate automatically when the outriggers are extended. Controls for the outriggers shall be mounted at the rear of the truck. A warning light shall be located on the driver's dashboard to provide an automatic warning when one or more of the outriggers have drifted out of the designed storage position.

3.11.11.6.1 Outrigger strength. The attachment points and components of the outriggers shall be of sufficient strength to carry the weight of the aerial ladder/water tower portions of the truck and live loads incident to firefighting operations without bending, twisting, fracture, or failure.

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3.11.11.7 Control station. A control station shall be provided at the base of the aerial ladder/water tower. The control station pedestal or panel shall be sized to accommodate the controls, gages, and switches specified in 3.11.11.7.1 through 3.11.11.7.9, and shall be located so that the operator's line of site is parallel to, or coincides with, the axis of the aerial ladder/water tower. The controls, gages, and switches on the control station shall be in accordance with the requirement for controls of MIL-STD-1472. Labels and instructions for the controls, gages, and switches on the control station shall be in accordance with the labeling requirements of MIL-STD-1472, and shall be provided with switch-controlled illumination of not less than 30 cd/m^2.

3.11.11.7.1 Hydraulic control. An aerial ladder/water tower hydraulic control that combines turntable rotation, ladder elevation and depression, and ladder extension and retraction by one-hand operation shall be mounted on the control station.

3.11.11.7.2 Water system control. A control handle for the 4-inch-dia master control valve shall be mounted on the control station beneath the gage specified in 3.11.11.7.3. The control shall be rotary.

3.11.11.7.3 Water pressure gage and flow meter. A water flow meter shall be mounted on the control station. The water meter shall have a range of 0 to not less than 1200 gpm. The meter face shall not be less than 3.5-inches dia. The gage shall be installed with a plate or collar, and shall be piped to the outlet side of the master control valve. The pressure gage shall have a range of not less than 0 to 600 psig.

3.11.11.7.4 Remote nozzle controls. Separate controls for the remote control nozzle shall be mounted on the control station in proximity to each other. The controls shall be:

a. Nozzle vertical control to elevate and depress the nozzle in a vertical plane.
b. Nozzle horizontal control to sweep the nozzle back and forth.
c. Stream-shaper control to adjust the nozzle tip.

3.11.11.7.5 Outrigger controls. A control to extend and retract the outriggers shall be mounted at the rear of the truck in a way that the operator may view all out rigger motion. The control shall be provided with an automatic means to render the outriggers inoperative (lockout) when the aerial ladder/water tower is elevated. A protected switch that unlocks the self-locking feature of the outriggers shall be mounted in proximity to the outrigger control. Protection for the switch shall completely surround the switch in such a manner that only intentional switching can be accomplished. A warning device shall be provided to automatically sound when the outriggers are in motion.

3.11.11.7.6 Master hydraulic electric switch. A master hydraulic electric switch that controls the electric power circuit to the 12-Vdc electric-motor-driven pump shall be mounted on the control station panel. The master hydraulic electric switch shall be equipped with a white-bezeled indicating light that will activate whenever the master hydraulic electric switch is in the "on" position.

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3.11.11.7.7 Throttle control switch. A throttle control switch shall be mounted on the control station panel and labeled in accordance with MIL-STD-1472. Activation of this switch by the operator shall prevent engine surging when minor hydraulic control movements are initiated. The throttle control switch shall be a spring-loaded, normally-closed type switch.

3.11.11.7.8 Hydraulic pressure gage. A hydraulic pressure gage shall be mounted on the control station panel. The gage face shall be color coded and provide pressure valves to indicate caution zones in yellow and danger zones in red.

3.11.11.8 Communication system. A communication system designed in conformance with MIL-STD-1472 speech transmission and reception requirements shall be provided between the control station and the top of the aerial ladder/water tower. The communication system shall allow clear communication between personnel at the control station and personnel at the remote nozzle controls at the top of the ladder/tower combination. An audio alerting signal, which is 20 dB above typical operating noise levels, shall be provided at both the control station and remote nozzle controls. This signal shall be button controllable at each end of the communication link.

3.11.11.9 Ladder bed locks. Ladder bed locks or safety lock outs, which will prevent action and movement of the aerial ladder/water tower while the truck is in motion, shall be installed, or built into the control system. Controls to lock the ladder/tower in running position and to release it for operation shall be automatic or provided at the control station panel. Ladder locking arms or similar locking devices when used shall be guarded so that their release motion is not a safety hazard to personnel.

3.11.11.10 Aerial ladder/water tower lighting. The aerial ladder/water tower shall have two sealed-beam lights mounted on either side of the base section of the ladder positioned to illuminate the ladder during night operations. In addition, a movable spotlight of not less than 100,000 candlepower shall be provided on the fly section end by which the monitor operator is enabled to observe the effect of the monitor stream. Both ladder illuminating lights and the spotlight shall be controlled by switches mounted on the control station panel.

3.11.11.11 Aerial ladder/water tower safety. The aerial ladder/water tower shall have the following safety features incorporated: A manual blocking plate or other devices shall be provided which can be installed around the ladder controls shall be provided to protect all ladder controls from accidental movement after the ladder has been positioned.

a. Automatic stops to prevent damage to the ladder/tower at full extension or full retraction shall be provided.
b. A means of locking the ladder/tower with rungs in alignment, with visible indication of the locking mechanism's locked or unlocked condition, shall be provided. When the rungs are locked in alignment, the retraction of the ladder/tower shall be prevented.

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c. A restriction device shall be incorporated into the ladder/tower operating mechanism, which will prevent retraction at a speed that will damage the ladder or endanger personnel in the event of a power failure.
d. A positive locking device shall be installed on the turntable that will prevent rotation of the turntable under any circumstances until the lock is released.
e. A load limiting device and automatic alarm that will prevent ladder/tower movement when the ladder/tower is loaded above the predetermined safe limits an instruction plate that specifies the safe load limit of the ladder/tower. If an instruction plate is used, it shall conform to 3.8.2.2.
f. Stops shall be provided to assure alignment of the base and fly sections during extension and retraction.
g. Indicators visible to the operator at the control station of the ladder/tower angle of elevation and alignment with the ladder's running position (ladder bed) shall be provided.
h. A lock-out device shall be installed to prevent movement of the ladder from the bedded position until all outriggers are in the recommended safe-operation position.

3.12 Physical protection.

3.12.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.12.1.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected, accessible position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to lubrication fittings shall be provided without removal or adjustment of accessories or parts. Panels or plates equipped with hand-operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

3.12.1.2 Enclosures with integral reservoir of lubricant. Enclosures such as engine crankcase, transmission housing, and gearcases which contain a reservoir of liquid lubricant shall be equipped with dipsticks marked in accordance with SAE J614, or sight gages to determine the acceptance level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining the lubricant. Drains shall be fitted with an SAE J531 or J532 magnetic drainplug. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant. Drainage shall be to a container on the ground when the truck is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the container. Accessibility to the drainplug, the filling means, and the

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lubricant-level checking device shall be provided without removal or adjustment of accessories or parts. Panels or plates equipped with hand-operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

3.12.2 Fungus and moisture-resistance. Electrical circuitry, including all components and connections, except as specified below, shall be protected from the effect of fungus growth and moisture penetration by an overall treatment with a varnish conforming to MIL-V-173, composition I.

a. Components or circuit elements which are inherently fungus and moisture-resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to provide a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.12.2.1 Varnish composition. When local air-pollution regulations preclude the use of MIL-V-173, composition I varnish, composition II varnish may be used for fungus and moisture-resistance. When composition II varnish is used, the contractor shall have evidence on hand that the use of composition I varnish is precluded and that the composition II varnish complies with Rule 442 of the South Coast Air Quality Management Control District.

3.12.3 Rustproofing. The truck shall be rustproofed in accordance with MIL-STD-1223.

3.13 Electromagnetic interference (EMI). Unless otherwise specified (see 6.2), the EMI emission and susceptibility characteristics of the truck shall be in accordance with
SAE J551.

3.14 Dimensions. The overall dimensions of the truck, including all lights, mounted equipment, and mirrors, shall not exceed a height of 138 inches, length of 456 inches, and a width of 108 inches.

3.15 Weight. The gvw of the truck, when loaded as specified in 3.7.1, shall be not more than the manufacturer's gvw ratings for the axles and tires.

3.16 Color. Those parts of the truck that are normally painted shall be gloss lime-yellow, color no. 13670 of FED-STD-595.

3.17 Finish. Those parts of the truck that are normally painted shall be gloss lime-yellow, color no. 13670, FED-STD-595. The cab and roof, down to the bottom of the windshield and ladder, shall be painted gloss white. The walk-surfaces and all treadplates shall be polished anodized aluminum. A white

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reflective tape band, not less than 10 inches wide, shall be affixed to the truck through the center line of the headlights. The tape shall be applied to all side surfaces over the full length of the truck except the pump panels. All bubbles and wrinkles shall be removed.

3.18 Identification marking. The truck shall be identified in accordance with MIL-STD-130. Identification name plates and parts data plates or labels shall be in accordance with MIL-STD-1223. The plates shall conform to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.18.1 Exterior markings. When specified (see 6.2), exterior markings on the truck shall be in accordance with MIL-STD-1223.

3.19 Workmanship.

3.19.1 Castings and forgings. All parts, components, and assemblies of the truck that include castings and forgings shall be free of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures that do not reduce mechanical properties or affect function.

3.19.2 Metal fabrication. Metal used in fabrication shall be free of kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel may be employed instead of shearing. All bends shall be made by controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.19.3 Bolted connections. Boltholes shall be accurately formed and shall have burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than five full threads.

3.19.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492, or Industrial Fastener Institute Standard No. 114 for break mandrel blind rivets.

3.19.5 Welders and welding.

3.19.5.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

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ANSI/ASME	Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1	Structural Welding Code, Section 5, Qualification.
ANSI/AWS D1.2	Structural Welding Code, Aluminum.

Certification shall be made available for review by the contracting officer or his designated representative. Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes, provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes.

3.19.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale, and other foreign matter that can be removed by chipping and wire brushing. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16-inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the truck shall be at the option of the contractor.

3.19.5.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be brushed with a stainless steel brush until free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert-gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or splatter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspections in section 4 are the minimum to be used to demonstrate compliance. Sampling in quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection require- ments specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the first article.

4.3.2 First article tests. The first article shall be subjected to the test marked "X" in column 1 of table I in the order indicated in the table. Failure of any test shall be cause for rejection of the first article.

TABLE I. Test schedule.

1st Art.	Qual. Conf.	Test	Req. para.	Test para.
1	2	3	4	5
X	X	Hydraulic system cleanliness	3.11.11.1.8	4.5.5.6
X	X	Hydraulic filters	3.11.11.1.4.1, 3.11.11.1.4.2	4.5.5.4
X	X	Hydraulic cylinder proof	3.11.11.1.7	4.5.5.5
X	X	Hydraulic pump contaminant tolerance	3.11.11.1.3	4.5.5.3
X		Hosebin and cabinet bottoms	3.11.6.1.1.1, 3.11.6.1.2, 3.11.6.2.3.1	4.5.2.11
X		Treadplate load	3.11.6.4	4.5.6
X		Exterior noise	3.9.1	4.5.2.4
X		Alternator output	3.11.1.7	4.5.2.8.1
X		Inverter voltage control	3.11.2.1.7	4.5.2.8.2
X		EMI	3.13	4.5.3
X	X	Mobility	3.7.1	4.5.2.2
X	X	Service brakes	3.11.1.6.1	4.5.2.7.1
X		Parking brakes	3.11.1.6.2	4.5.2.7.2
X	X	Engine brake	3.11.1.6.3	4.5.2.7.3
X	X	Pumping capacity	3.7.2	4.5.2.3
X		Fuel system capacity	3.11.1.2.3	4.5.2.6
X		Pump drive temperature	3.11.5.1.1	4.5.2.3.1
X		Discharge relief valve	3.11.5.3.2	4.5.2.3.2
X		Foam system	3.11.4	4.5.2.9
X		Pumping stability	3.11.11	4.5.4

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TABLE I. Test schedule. (cont'd)

1st Art.	Qual. Conf.	Test	Req. para.	Test para.
1	2	3	4	5
X		Altitude pumping (see 6.2)	3.10.1	4.5.2.5.1
X	X	Hydrostatic	3.11.5	4.5.2.10
X		Throttle control switch	3.11.11.7.7	4.5.5.8
X		Hydraulic backup	3.11.11.1	4.5.5.1
X		Hydraulic pump inlet pressure	3.11.11.1.2.4	4.5.5.2
X		Hydraulic external leakage	3.11.11.1.9	4.5.5.7
X		Cold starting and operating	3.10.2	4.5.2.5.2
X		Hot starting	3.10.3	4.5.2.5.3
X		Maintainability	3.6	4.5.2.1

4.4 Quality conformance inspecition.

4.4.1 Sampling. Sampling for examinations and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each truck shall be examined for the critical defects specified in 4.5.1. Presence of any defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordnace with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with MIL-STD-105 shall be subjected to the tests marked "X" in column 2 of table 1. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The truck shall be examined as specified herein for the following defects:

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Critical

Defect no.	Defect	Requirement paragraph
1.	Master pressure gage not compound type or with dial face size less than specified.	3.11.6.3.2.1
2.	Master suction gage not compound type or with dial face size less than specified.	3.11.6.3.2.2
3.	Backup warning not as specified.	3.11.8.4
4.	Truck's gvw exceeds gvw ratings for axles and tires.	3.15

Major

Defect no.	Defect	Requirement paragraph
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
104.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.1.2
105.	Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the truck.	3.3.2
106.	Design of the truck not as specified.	3.4
107.	Angle of departure less than specified in NFPA Code No. 1901.	3.4.1
108.	Construction of the truck not as specified.	3.5
109.	Interchangeability of parts not as specified.	3.5.1
110.	Threads of the piping not in conformance with FED-STD-H28.	3.5.2
111.	Wiring not identified; identification not consistent; protection, installation, and attachment not as specified; or fuses used or circuit breakers omitted.	3.5.3
112.	Mounting not secured to the truck as specified.	3.5.4

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Major

Defect no.	Defect	Requirement paragraph
113.	Attaching chains not in conformance to RR-C-271, type II, class 6, corrosion resistant or zinc coated or not attached as specified.	3.5.5
114.	Major components not removable and replaceable or access openings not sized, finished, and covered as specified.	3.6
115.	Human factors not as specified.	3.8.1
116.	Control labels not in accordance with MIL-STD-1472 or universal symbology for driver's gages, instruments, and controls not in accordance with SAE J1500.	3.8.2.1
117.	Identification and instruction plates not installed or not of material specified.	3.8.2.2
118.	Nameplate marking not as specified.	3.8.2.3
119.	Personnel protection not as specified or protective devices impair operating efficiency of the truck.	3.9.2
120.	Boarding and riding protection not as specified.	3.9.3
121.	Chassis not front-wheel-steer, rear-wheel-drive type.	3.11.1
122.	Frame not heavy-duty fire service type or altered so as to reduce its strength.	3.11.1.1
123.	Engine selection, accessories, or type not as specified.	3.11.1.2
124.	Engine not 370 bhp at 2600 rpm, or engine not EPA certified.	3.11.1.2.1
125.	Contractor does not have bhp certification available.	3.11.1.2.1.1
126.	Engine lubrication system not as specified.	3.11.1.2.2
127.	Engine fuel system filler opening, cap position, fuel line installation, or tank position not as specified.	3.11.1.2.3
128.	Engine cooling system not of capacity recommended or not of materials specified.	3.11.1.2.4
129.	Engine starting system allows engine start with transmission in other than neutral position or ignition switch is not keyless with a pilot light.	3.11.1.2.5
130.	Engine exhaust system does not include components specified or is not installed as specified.	3.11.1.2.6

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Major

Defect no.	Defect	Requirement paragraph
131.	Any component of drive assembly with a torque capacity rating less than the maximum torque developed by the engine.	3.11.1.3
132.	Transmission not fully automatic, 4-speed, with reverse shift protection and neutral locking device.	3.11.1.3.1
133.	Drive axle not full-floating type.	3.11.1.3.2
134.	Springs not of rated capacity.	3.11.1.4.1
135.	Shock absorbers not double-acting, hydraulic type or not matched to suspension capacity.	3.11.1.4.2
136.	Wheels and tires do not conform to TRA requirements for truck gvw at 50 to 60 mph.	3.11.1.5
137.	Front and rear wheels not type specified or not inherently in balance.	3.11.1.5.1, 3.11.1.5.2
138.	Tires not at least 100-level quality, not sized for load, or with wrong tread pattern.	3.11.1.5.3
139.	Tire valves, caps, and cores not as specified.	3.11.1.5.3.1
140.	Spare wheel and tire not furnished or as specified.	3.11.1.5.4
141.	Brakes do not conform to latest DOT requirements.	3.11.1.6
142.	Service brake mechanisms not accessible for adjustment, or service brake lines not installed and protected as specified.	3.11.1.6.1
143.	Parking brake not all-wheel locking, not type specified, or without emergency release mechanism.	3.11.1.6.2
144.	Engine brake not of type specified.	3.11.1.6.3
145.	Generating system has alternator of less than 145-amp output at 12 Vdc.	3.11.1.7
146.	Steering system not hydraulic powerassisted, or provides a cramping angle of less than 28 degrees.	3.11.1.8
147.	Rear step not constructed to serve as bumper or with riding surface more than 25.5 inches above level ground.	3.11.1.9.1
148.	Front bumper not heavy duty unit attached to frame.	3.11.1.9.2
149.	Towing hook or towing lugs do not permit attachment of a medium-duty towing bar, or reduce the angle of departure.	3.11.1.10

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Major

Defect no.	Defect	Requirement paragraph
150.	Towing devices not as specified.	3.11.1.10.1
151.	Towing lugs not as specified.	3.11.1.10.2
152.	Partition in cab does not have sliding glass window or window is less than 50 percent of windshield area.	3.11.2
153.	Driver's compartment not separated from engine by insulation on firewall or not finished, trimmed, and sealed as specified.	3.11.2.1
154.	Driver's compartment seats not upholstered as specified, or undersized; driver's seat not adjustable.	3.11.2.1.1
155.	Driver's dashboard does not extend full width of cab, lacks specified depth, or permits dash lighting to reflect on windshield.	3.11.2.1.2
156.	Driver's gages and instruments not installed as specified, are fewer than minimum required, or are not lighted by head-light switch control.	3.11.2.1.2.1
157.	Driver's operating controls not installed as specified.	3.11.2.1.2.2
158.	Gage and instrument monitors not as specified.	3.11.2.1.2.3
159.	Glass and glazing not of quality or not sealed as specified.	3.11.2.1.3
160.	Windshield wiping and cleaning system not in conformance with SAE J198 or does not have reservoir of at least 2-quart capacity.	3.11.2.1.4
161.	Heater in driver's compartment not as specified.	3.11.2.1.5
162.	Transmission shift-control pedestal quadrant or location not as specified.	3.11.2.1.6
163.	Inverter not equipped with switch and voltmeter, or not located as specified.	3.11.2.1.7
164.	Communications compartment omitted, not of size specified, or not adapted to communications equipment.	3.11.2.1.8
165.	Rearview mirrors not correct type, quantity or mounting, or stick-on convex mirror used.	3.11.2.1.9.1
166.	Seatbelts and anchors not as specified or without correct buckles and retractors.	3.11.2.1.9.2
167.	Interior lights not correctly located or not correctly switched.	3.11.2.1.9.3

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Major

Defect no.	Defect	Requirement paragraph
168.	Dual sun visors not installed in driver's compartment.	3.11.2.1.9.4
169.	Jump-seat area not separated by engine cover.	3.11.2.2
170.	Engine cover not ventilated or ventilating louvers send engine compartment air toward jump-seat occupants.	3.11.2.2.1
171.	Jump seats not sized, upholstered, or bracketed as specified.	3.11.2.2.2
172.	Jump-seat safety bars not size, type, or locked as specified.	3.11.2.2.2.1
173.	Jump-seat dome light omitted or without control switch.	3.11.2.2.2.2
174.	Soft suction hose compartment undersized, with projections inside, without drain holes, or not finished to avoid chafing hose.	3.11.2.3
175.	Water tank not mounted as specified.	3.11.3
176.	Water tank size, shape, baffling, top, or top-fastening not as specified.	3.11.3.1
177.	Water tank not manufactured of correct materials or not cleaned as specified before installation.	3.11.3.2
178.	Water tank fill opening less than 64 square inches, unscreened, without specified closure, or overflow drain provisions as specified.	3.11.3.3
179.	Water tank outlet not as specified.	3.11.3.4
180.	Foam system not of type or capacity specified or not of corrosion-resistant materials.	3.11.4
181.	Foam-liquid tank not removable or not located near fire pump.	3.11.4.1
182.	Foam-liquid tank not of 30-gallon capacity or not baffled to prevent excessive surging.	3.11.4.1.1
183.	Foam-liquid tank not manufactured of correct materials or not cleaned as specified before installation.	3.11.4.1.2
184.	Foam-liquid tank filler opening not sized, screened, or covered as specified.	3.11.4.1.3
185.	Foam-liquid tank not vented or with vent piping not as specified.	3.11.4.1.4
186.	Foam-liquid tank outlet not screened.	3.11.4.1.5

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Major

Defect no.	Defect	Requirement paragraph
187.	Foam-liquid tank not equipped with drain, drain valve, or drain pipe directed as specified.	3.11.4.1.6
188.	Foam-liquid metering valve not as specified.	3.11.4.2
189.	Foam system valving and piping not as specified.	3.11.4.3
190.	Foam system flushing provisions omitted or do not provide for complete flushing of system.	3.11.4.3.1
191.	Foam system controls not as specified or without instruction plate.	3.11.4.4
192.	Fire pump not UL rated or priming device does not conform to UL 822.	3.11.5.1
193.	Suction section not of corrosion-resistant metal or not galvanized as specified.	3.11.5.2
194.	Fire pump suction manifold not as specified.	3.11.5.2.1
195.	Large suction inlets not of correct size or location; without strainers; or not valved or capped as specified.	3.11.5.2.2
196.	Small suction inlets not of size, type, or location specified or not plugged as specified.	3.11.5.2.3
197.	Water tank suction piping without shutoff and check valves.	3.11.5.2.4
198.	Suction manifold fittings at ports not corrosion-resistant.	3.11.5.2.5
199.	Discharge section not of corrosion-resistant metal or not galvanized as specified.	3.11.5.3
101a.	Fire pump discharge manifold not as specified.	3.11.5.3.1
102a.	Indicator light provision not incorporated into pressure relief valve.	3.11.5.3.2
103a.	Discharge control valves not of type specified, or not numbered and tagged as specified.	3.11.5.3.3
104a.	Discharge outlets not of correct quantity, size or location, or not labelled as specified.	3.11.5.3.3
105a.	Discharge piping not as specified.	3.11.5.3.5
106a.	Discharge outlet drains of wrong size, improperly located, or with drain lines not as specified.	3.11.5.3.6

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Major

Defect no.	Defect	Requirement paragraph
107a.	Truck body does not provide for attachment of storage units or is not reinforced at points of attachment.	3.11.6
108a.	Hosebins not integral with the truck body.	3.11.6.1
109a.	Small-hose bin not located, sized, divided, and with smooth interior.	3.11.6.1.1
110a.	Any hosebin bottoms not removable; any hosebin or cabinet bottoms without raised crossmembers with attachments for floors, or without drain holes between the cross-members.	3.11.6.1.1.1, 3.11.6.1.2, 3.11.6.2.3.1
111a.	Any hosebin or cabinet floors not of materials specified, with slat sizes or slat spacing not as specified, or not fastened to botton crossmembers by counter-sunk screws or bolts.	3.11.6.1.1.2, 3.11.6.2.3.1
112a.	Large-hose bin not located, sized, divided, and with smooth interior, or without removable bottom and floor.	3.11.6.1.2
113a.	Attached equipment cabinet arrangement not as specified.	3.11.6.2
114a.	Equipment cabinet height, depth, or recess less than specified; filler panels or skirts omitted; tops and fronts not flush horizontally and vertically, without gaps.	3.11.6.2.1, 3.11.6.2.2
115a.	Any cabinet doors not as specified.	3.11.6.2.3.2
116a.	Any cabinet illumination omitted or illumination not controlled through door switches.	3.11.6.2.3.3
117a.	Any cabinet ventilation and drainage omitted.	3.11.6.2.3.4
118a.	Cabinet mountings not adequate to support three fire fighters without fracture, warp, or sag.	3.11.6.2.3.5
119a.	Pump operator's panel not located, sized, labelled, or arranged as specified.	3.11.6.3
120a.	Pump operator's control installed without collars or plates.	3.11.6.3.1
121a.	Engine throttle control not screw-locking and center-pushbutton release type.	3.11.6.3.1.1
122a.	Heat exchanger control not as specified on pump operator's panel, when heat exchanger is provided.	3.11.6.3.1.2

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Major

Defect no.	Defect	Requirement paragraph
123a.	Foam-liquid control not of type specified, without dial and pointer indicator, or without instruction plate.	3.11.6.3.1.3
124a.	Water tank shutoff control not linear or does not have adjacent drain control.	3.11.6.3.1.4
125a.	Small suction inlet controls not located as specified or not linear.	3.11.6.3.1.5
126a.	Discharge valve controls not linear, not symmetrically grouped, or not numbered as specified.	3.11.6.3.1.6
127a.	Fire pump drive control not linear and locking type.	3.11.6.3.1.7
128a.	Discharge relief valve control not rotary type, without dial and pointer indicator, or without indicating light.	3.11.6.3.1.8
129a.	Discharge outlet drain controls not mounted beneath associated discharge valve control.	3.11.6.3.1.9
130a.	Pump operator's gages installed without plates or collars.	3.11.6.3.2
131a.	Pressure gages for each discharge outlet not of type specified, not properly sized, or not grouped symmetrically with associated controls.	3.11.6.3.2.3
132a.	Engine and transmission gages not provided, not sized and located, or without indicator light as specified.	3.11.6.3.2.4
133a.	Water tank gage not installed or not located as specified.	3.11.6.3.2.5
134a.	Pump operator's instruments installed without plates or collars.	3.11.6.3.3
135a.	Engine tachometer on pump operator's panel not interchangeable with tachometer on dashboard, not in proximity to throttle, or without means of checking.	3.11.6.3.3.1
136a.	Voltmeter missing, not properly sized, or not located as specified on pump operator's panel.	3.11.6.3.3.2
137a.	Test connections missing, not properly sized or threaded, not located and labelled, or not plugged as specified.	3.11.6.3.3.3
138a.	Pullout treadplate not sized or locked as specified.	3.11.6.4

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Major

Defect no.	Defect	Requirement paragraph
139a.	Electrical wiring for 110-vac circuits not separate from 12-vdc circuits.	3.11.7
140a.	Electrical equipment not in conformance with requirement 1 of MIL-STD-454, not protected as specified, or 6-v equipment incorporated in truck.	3.11.7.1
141a.	Batteries not as specified.	3.11.7.2
142a.	Battery tray(s) not ventilated or treated and battery terminal guards omitted.	3.11.7.2.1
143a.	Battery circuits do not conform to the appendix to NFPA Code No. 1901.	3.11.7.3
144a.	Battery charging circuits, station not as specified.	3.11.7.3.1
145a.	Station battery-charging provisions omitted or polarized receptacle not located as specified.	3.11.7.3.2
146a.	Battery charging circuits on truck not protected with reverse polarity device or not switched as specified.	3.11.7.3.2
147a.	Truck operating lights not as specified.	3.11.7.4.1
148a.	Equipment operating lights not as specified.	3.11.7.4.2
149a.	Warning lights do not conform to NFPA Code No. 1901 or are not as specified.	3.11.7.4.3
150a.	Floodlights not as specified.	3.11.7.4.4
151a.	Floodlight receptacles not of type, location, or wired as specified.	3.11.7.4.5
152a.	Lighting system controls not labelled in accordance with MIL-STD-1472.	3.11.7.5
153a.	Headlight control not of type specified.	3.11.7.5.2
154a.	Warning light controls not of type, location, or not shielded to prevent glare.	3.11.7.5.2
155a.	Floodlight receptacle controls omitted.	3.11.7.5.3
156a.	Siren not of type specified; not wired to connect to radio; not located and controlled; or without brake as specified.	3.11.8.1
157a.	Horns not of type, location, or controlled as specified.	3.11.8.2
158a.	Buzzer omitted or not switched as specified.	3.11.8.3

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Major

Defect no.	Defect	Requirement paragraph
159a.	Handrails and grabrails not of size, mounting, or locations as specified.	3.11.9
160a.	Stowed component-attaching devices not positive locking, quick-release type.	3.11.10
161a.	Soft suction hose not of length, diameter, or quality specified.	3.11.10.1
162a.	Reducer not of size, type, or not stowed as specified.	3.11.10.2
163a.	Ladders do not conform to NFPA Code No. 1931 or not of types specified.	3.11.10.3
164a.	Ladders not stowed where specified, not padded, or not mounted in specified devices.	3.11.10.3.1
165a.	Axes not of types specified, or not mounted where specified.	3.11.10.4
166a.	Pike pole not of type or length specified or not stowed where specified.	3.11.10.5
167a.	Wheel chocks not of quantity specified or not stowed where specified.	3.11.10.6
168a.	Hard suction hose not as specified.	3.11.10.7
169a.	Hard suction hose storage not as specified.	3.11.10.7.1
170a.	Strainer not as specified or not stowed as specified.	3.11.10.8
171a.	Aerial ladder/water tower not 100 ft long.	3.11.11
172a.	Hydraulic system not of materials compatible with hydraulic oils specified.	3.11.11.1
173a.	Hydraulic fittings not of type specified for port size applicable.	3.11.11.1.1
174a.	Hydraulic lines not installed as specified.	3.11.11.1.2
175a.	Pressure hoses not of type or sizes specified, not pressure rated to or above relief setting, or bent beyond minimum radius.	3.11.11.1.2.1
176a.	Pressure tubing not of type or sizes specified, burst pressure less than specified, or improperly installed.	3.11.11.1.2.2
177a.	Pressure tube fittings have a burst pressure less than specified.	3.11.11.1.2.3
178a.	Suction lines without flexible characteristics or hose and clamps not as specified.	3.11.11.1.2.4

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Major

Defect no.	Defect	Requirement paragraph
179a.	Return lines not as specified.	3.11.11.1.1.5
180a.	Hydraulic system filters omitted.	3.11.11.1.4
181a.	Return line and pressure line filters not removable without spilling or draining oil, or not equipped with provisions for remote contamination indication.	3.11.11.1.4.1 3.11.11.1.4.2
182a.	Hydraulic fluid reservoir not designed, constructed, or installed as specified.	3.11.11.1.5
183a.	Pressure relief valve not set to maximum system pressure.	3.11.11.1.6
184a.	Hydraulic cylinders not manufactured as specified.	3.11.11.1.7
185a.	Turntable not of dimension specified.	3.11.11.2
186a.	Aerial ladder/water tower not constructed as specified.	3.11.11.3
187a.	Aerial ladder/water tower system not as specified.	3.11.11.4
188a.	Master control valve not of type or material specified.	3.11.11.4.1
189a.	Remote control nozzle not of size specified, does not have vertical and horizontal sweep functions separate, or does not have duplicate controls as specified.	3.11.11.5
190a.	Remote nozzle protection not as specified.	3.11.11.5.1
191a.	Outriggers not equipped with automatically activated flashing lights.	3.11.11.6
192a.	Controls, gages, and switches on control station not arranged, labelled, or illuminated as specified.	3.11.11.7
193a.	Ladder/tower hydraulic control not one-hand type.	3.11.11.7.1
194a.	Control handle of master control valve not located as specified.	3.11.11.7.2
195a.	Water pressure gage not of range and size specified.	3.11.11.7.3
196a.	Remote nozzle and stream-shaper controls not mounted together.	3.11.11.7.4
197a.	Outrigger control not equipped with lockout or unlocking switch not protected as specified.	3.11.11.7.5
198a.	Master electric switch not equipped with indicator light.	3.11.11.7.6
199a.	Engine starter switch not on control station.	3.11.11.7.8

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Major

Defect no.	Defect	Requirement paragraph
101b.	Hydraulic pressure gage not on control station.	3.11.11.7.9
102b.	Communication system between control station and top of ladder/tower omitted.	3.11.11.8
103b.	Ladder bed locks not as specified.	3.11.11.9
104b.	Ladder/tower lights not of type, location, or controlled as specified.	3.11.11.10
105b.	Ladder/tower safety provisions not as specified.	3.11.11.11
106b.	Truck not serviced as specified or lubrication data missing.	3.12.1
107b.	Lubrication fittings not of type, location, or accessible as specified.	3.12.1.1
108b.	Enclosures of integral lubricant reservoirs not as specified.	3.12.1.2
109b.	Fungus-and-moisture treatment not as specified.	3.12.2, 3.12.2.1
110b.	Truck not rustproofed in accordance with MIL-STD-1223, when specified.	3.12.3
111b.	Overall dimensions of truck greater than specified.	3.14
112b.	Finish cleaning, treatment, and painting not as specified.	3.17
113b.	Bolted connections not as specified.	3.19.3
114b.	Riveted connections not as specified.	3.19.4
115b.	Certification of welders not available.	3.19.5.1
116b.	Steel welding not as specified.	3.19.5.2
117b.	Aluminum welding not as specified.	3.19.5.3

Minor

201.	Color not as specified.	3.16
202.	Identification marking omitted or not as specified.	3.18
203.	Exterior markings not as specified.	3.18.1
204.	Castings and forgings not free of extraneous materials.	3.19.1
205.	Metal fabrication not as specified.	3.19.2

4.5.2 Tests.

4.5.2.1 Maintainability test. After all tests involving the use of the truck engine have been completed, perform the maintenance operations specified in 3.6a through 3.6i. The engine-driven belts, alternator, coolant hoses, battery cables, batteries, and starter shall be inspected upon removal and shall be reinstalled instead of being replaced if inspection indicates a satisfactory

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operating condition of the components involved. Screens and strainers shall be cleaned and inspected and may be reinstalled if inspection indicates a satisfactory operating condition of the screen or strainer. Fluids and filters shall be replaced. Any maintenance operation performed that exceeds the time allocated in 3.6 for that operation, including inspection and cleaning of the component involved, shall constitute failure of this test.

4.5.2.2 Mobility test. Prepare the truck for testing as follows:

a. Fill the water tank with not less than 200 gal of water.
b. Fill the foam-liquid tank with not less than 30 gal of foam-liquid conforming to MIL-F-24385.
c. Fill the fuel tank to capacity with diesel fuel.
d. Fill all lubricant and hydraulic reservoirs. Lubricants shall be those indicated on a lubrication data plate or lubrication decals.
e. Load the truck with an evenly distributed, simulated load of 1500 lbs.
f. Load the truck with a driver, a passenger, and simulated deadweights of not less than 175 lbs each in each jump seat and on the rear platform. If the driver and passenger's collective weights do not equal or exceed 350 lbs, a deadweight load equal to the difference between their collective weights and 350 lbs shall be placed on the cab floor on the passenger's side. Secure the simulated loads by a positive means that will prevent load shifting or loss of any part of the load during running, turning, or braking during the test. When the truck is fully prepared for testing, start the engine and allow it to operate for a sufficient time to bring the engine coolant temperature up to the engine manufactur-er's recommended operating temperature. When engine coolant temperature rise is sufficient, operate the truck over a test course of 100 miles of paved highway and 50 miles of graded dirt road. The test course shall be negotiated at maximum posted speeds up to 55 mph on paved highway and up to 30 mph on graded dirt road. Nonconformance of the truck to any performance ormobility requirement of NFPA Code No. 1901 shall constitute failure of this test.

4.5.2.3 Pumping capacity test. The pumping capacity test conditions are an ambient air temperature of not less than 125 ^{0}F and an altitude of 2000 ft above sea level. These test conditions may be actual or simulated. Simulated test conditions shall be approved before testing by the contracting officer or the designated representative. The test shall consist of testing the priming device operation, the suction of the pump, and 3 hours of actual pumping time. The truck shall be prepared by attaching 20 ft of 6-inch suction hose to the 6-inch suction inlet. The suction hose shall be terminated with a strainer as specified in 3.11.10.8. Place the strainer end of the hose in a reservoir or similar body of water that will permit the withdrawal of not less than 5,000 gal without drafting mud or silt. Attach a calibrated suction gage and a calibrated pressure gage equivalent to the gages specified in 3.11.6.3.2.2 and 3.11.6.3.2.1, respectively, to the test connections on the pump operator's panel. Attach hoses to outlets on the truck and direct their flows back to the water source. Fill the fuel tank to its maximum capacity. Start the truck engine, engage the pump drive, and bring the engine up to the recommended rpm for priming that is recommended by the priming device manufacturer. Operate the priming device and observe the test suction gage to determine that suction has been established.

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Increase engine speed and open the outlet control valves until the truck is discharging 1000 gpm at 150 psig as observed on the calibrated discharge test gage. Continue pumping at this rate for not less than 2 hours. After 2 hours, adjust outlet control valves and engine speed to pump 700 gpm at 200 psig as observed on the calibrated discharge test gage. Continue pumping at this rate for not less than 30 minutes. After 30 minutes, adjust outlet control valves and engine speed to pump 500 gpm at 50 psig as observed on the calibrated discharge test gage. Continue pumping at this rate for another 30 minutes. Failure of the priming device to operate or establish suction within 30 seconds, or failure of the truck to establish and maintain the discharge rates at the pressures specified, shall constitute failure of this test.

4.5.2.3.1 Pump drive temperature test. During the pumping capacity test specified in 4.5.2.3, a thermocouple and calibrated temperature readout device shall be connected into the pump drive gear-oil reservoir. The gear-oil temperature shall be monitored on the readout device throughout the entire 3 hours of the pumping capacity test. A gear-oil temperature that exceeds the pump drive manufacturer's recommended maximum gear-oil temperature shall constitute failure of this test.

4.5.2.3.2 Discharge relief valve test. At the end of the pumping capacity test, establish a discharge rate of 1000 gpm at 75-psig pressure. Raise the discharge relief valve setting to not more than 85 psig. Slowly close the control valves on the discharging lines while observing the test pressure gage. Open the control valves on the discharging lines and reset the pressure relief valve to 160 psig. Increase the discharge pressure to 150 psig while maintaining a discharge rate of 1000 gpm. Slowly close the control valves on the discharging lines while observing the test pressure gage. A pressure reading increase of more than 30 psig above the discharge pressure established before the control valves on the discharging lines were started to close, through full closing of the valves, shall constitute failure of this test.

4.5.2.4 Noise-level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, and one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with 3.9.1 provisions shall constitute failure of this test.

4.5.2.5 Environmental tests.

4.5.2.5.1 Altitude pumping test. Any truck ordered for a service altitude over 2000 ft above sea level shall be tested as specified in 4.5.2.3, except that the test conditions shall be an ambient air temperature of 75 ^{0}F $\pm$ 10 ^{0}F and an altitude equal to the service altitude ordered. Failure of the test shall be as specified in 4.5.2.3.

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4.5.2.5.2 Cold starting and operating test. Cold soak the truck for a period of not less than 24 hours in an ambient air temperature of not greater than -25 ^{0}C. Start the truck and begin operating (driving) it within 30 seconds of start. Any failure of the engine to start within 30 seconds, engine stalling after commencement of operation, or failure of the transmission to shift into appropriate gear ranges shall constitute failure of this test.

4.5.2.5.3 Hot starting test. Conduct the test in an ambient air temperature of not less than 125 ^{0}F. Start and operate the truck engine until the engine coolant temperature reaches to the engine manufacturer's recommended operating temperature. Stop the engine and wait for 2 minutes. After the 2-minute wait, restart the engine. Failure of the engine to restart shall constitute failure of this test.

4.5.2.6 Fuel system capacity test. Measure the fuel tank capacity. Before starting the pumping test specified in 4.5.2.3, fill the fuel tank to capacity and record the elapsed time indicated by the engine hourmeter to a tenth of an hour. Upon completion of the pumping test, refill the fuel tank to capacity. Record the gal required to refill the tank to the nearest tenth of a gal and record the time indicated on the engine hourmeter to a tenth of an hour. Subtract the pre-test recorded engine-hour time from the post-test recorded engine-hour time. Divide the recorded gal required to refill the tank by the remainder to obtain the average gal per hour used. Add the average gal per hour used to the number of gal required to refill the tank to obtain a sum that represents the fuel tank capacity. Nonconformance to 3.11.1.2.3 shall constitute failure of this test.

4.5.2.7 Brake tests. The truck, loaded as specified, shall be subjected to the brake test specified by SAE J992 for compliance to SAE requirements for the size vehicle furnished.

4.5.2.7.1 Service brake test. During or immediately after the mobility test specified in 4.5.2.2, and while the truck is still loaded, drive the truck up a grade that is not less than 30 percent. When the truck is completely on the grade, bring it to a full stop on the grade and hold it there with only the service brake for 1 minute. Release the service brake and drive the truck onto a paved, dry, level surface that is free of loose material. Bring the truck speed to 20 mph and apply the service brakes in a manner that simulates an emergency stop. Measure the distance between the point where the service brakes were applied and the point where the truck stopped, then resume driving. Drive the truck down the same grade it had been driven up earlier in the test but bring it to a full stop on the grade. Hold the truck on the grade with only the service brake for one minute. Any drifting of the truck from the full-stop position, either up or down while stopped on the grade, or failure to come to a full stop within 30 ft on the level surface while applying no more than 75 lbs of force on the brake pedal, shall constitute failure of this test.

4.5.2.7.2 Parking brake test. Connect a calibrated air pressure gage to the truck's air tank. Start the truck engine and bring the tank's air pressure up to not less than 100 psig. Pull the parking brake completely "on" and leave the control in its fully extended position. Shutoff the truck engine. Observe and record the air-pressure reading on the calibrated pressure gage. Leave the

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parking brake "on" for not less than 24 hours. After 24 hours, observe and record the air pressure reading on the calibrated air-pressure gage. A drop in air pressure in the air tank of 25 lbs or more after 24 hours shall constitute failure of this test.

4.5.2.7.3 Engine brake test. Operate the truck over a 5-mile-long segment of paved highway at posted speeds up to 55 mph. At two different intervals, turn the engine brake switch to the "on" position. During the intervals when the switch is "on", remove pressure from the accelerator and observe if the truck speed is immediately retarded. Failure to immediately retard truck speed when pressure is removed from the accelerator while the engine brake switch is in the "on" position, or any immediate retarding of the truck speed that is not attributable to normal engine compression when the engine brake switch is in the "off" position, shall constitute failure of this test.

4.5.2.8 Generating system tests.

4.5.2.8.1 Alternator output test. Start the truck engine and increase the engine speed to 1000 rpm. Use an ammeter to measure the alternator/voltage regulator output in amps at this engine speed. Then slow the engine to the engine manufacturer's recommended idle speed and use the ammeter to measure the inverter output in amps at the idle speed. An alternator/voltage regulator output of less than 70 amps at an engine speed of 1000 rpm or an inverter output of less than 17.40 amps at engine idle speed shall constitute failure of this test.

4.5.2.8.2 Inverter voltage control test. Measure the inverter voltage and cyclic output under inductive loads of 2000, 4000 and 6000 W. Any voltage reading that is less than 105 Vac or more than 115 Vac, or any cyclic output that varies by more than 1 Hz from 60 Hz, shall constitute failure of this test.

4.5.2.9 Foam system test. Fill the foam-liquid tank to capacity, using either foam-liquid or a dyed-water substitute. Connect a hose to a discharge outlet that will accommodate a flow of not less than 200 gpm. Open the appropriate discharge valve. Connect a suction hose to an appropriate suction inlet. Start the engine, engage the fire pump drive, prime, and establish suction and a discharge flow of not less than 200 gpm. Set the foam-liquid metering valve or proportioner to deliver a 3-percent solution. Open the foam-liquid shutoff valve for exactly 1 minute and then close the valve. Refill the foam-liquid tank to capacity. Measure and record the amount of liquid required. Reset the foam-liquid shutoff valve for exactly 1 minute and then close the valve. Refill the foam-liquid tank to capacity. Measure and record the amount of liquid required. Drain the foam-liquid tank and flush and clean the entire foam system. Failure of the foam system to have discharged not less than 6 gal of liquid in solution in 1 minute at the 3-percent setting or not less than 12 gal of liquid in solution in 1 minute at the 6-percent setting shall constitute failure of this test.

4.5.2.10 Hydrostatic test. Prepare the truck for testing by emptying the water and foam-liquid tanks, and cleaning, if necessary, the foam piping and controls. Cap or plug all section inlets and all discharge outlets on the entire truck with valves that will facilitate bleeding of air from the fire pump and

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piping system. Discharge lines to the remote control nozzle may be plugged by closing the nozzle control. Close all drain valves and fully open all control valving between the fire pump and the suction inlets and discharge outlets. Connect calibrated pressure gages to both female test connections on the pump operator's panel. Connect the external water-pumping system to one of the small suction inlets. Start the external pumping system and charge the fire pump and piping system while bleeding all air out of the system. Bring the water pressure in the fire pump up to not less than 350 psig on both gages and maintain that pressure for not less than 10 minutes. Inspect the fire pump case, manifolds, control and drain valves, and all piping for evidence of leaks. Inspect the interior of the water and foam-liquid tanks for evidence of water indicating leakage past the tank to pump valve. Any evidence of leakage in or around the pump, manifolds, control or drain valves, piping and fittings or check valves shall constitute failure of this test.

4.5.2.11 Hosebin and cabinet bottom test. Remove the floors from each hosebin and cabinet before testing. With the floor removed, chalk-mark a square ft area on the bottom that does not incorporate a crossmember. Place a 1-inch-thick, one-square-foot piece of wood on the chalk-marked area. Place a static load of not less than 200 lb on the wood. After 10 minutes, remove the test load and wood. Examine the bottom for cracks or deformation or a failure to return to original flatness when the load is removed. Evidence of fracture, cracks or any permanent deformation shall constitute failure of this test.

4.5.3 EMI test. When conformance to EMI emission and susceptibility has been specified (see 6.2), the truck shall be tested in accordance with the test requirements of SAE J551. Nonconformance to SAE J551 shall constitute failure of this test.

4.5.4 Pumping stability test. Prepare the truck for testing as specified in 4.5.2.3, except that the ambient air temperature may be any temperature above +32 °F; the altitude may be any altitude above sea level; and the discharge of water shall be through the remote control nozzle on the end of the ladder/tower only. Use the truck engine to power the hydraulic system and extend the outriggers until they are locked into position to stabilize the truck. Elevate the ladder/tower from its riding position and extend it to its maximum extended condition. Rotate the ladder/tower to a heading of 35 degrees off the heading of the truck and depress it to 2 degrees below the horizontal. Engage the fire pump drive and prime and establish suction. Open the master control valve and adjust engine speed and the remote control nozzle until a flow of 1000 gpm at 150 psig is discharging through the remote control nozzle in a horizontal direction away from the truck. Release the hydraulic controls for not less than 2 minutes. After 2 minutes, elevate the ladder/tower to not less than 75 degrees and simultaneously adjust the elevation of the remote control nozzle to continue the discharge in a horizontal direction away from the truck. Release the hydraulic controls while the ladder/tower is in this elevation for not less than 2 minutes. Repeat the cycles of depressing and elevating the ladder/tower, with release of the hydraulic controls at the depressed and elevated angles specified on headings of 90 degrees and 140 degrees from the heading of the truck. The discharge rate shall be maintained continuously throughout all movements of the ladder/tower. After the last cycle, discontinue discharging by closure of the master valve and lowering of engine speed. Depress the fully extended ladder/tower to horizontal

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while still on the 140-degree heading away from the truck heading. Secure a 250-lb load on the ladder in a position most likely to be occupied by personnel operating the remote control nozzle with the controls near the nozzle. Elevate the ladder/tower and load to the horizontal. Open the master control valve, adjust engine speed and the remote control nozzle to reinstate a discharge of 1000 gpm at 150 psig through the nozzle in a horizontal direction away from the truck. Release the hydraulic controls for not less than 2 minutes and then elevate the ladder/tower and adjust the remote control nozzle to attain an elevation of 75 degrees above the horizontal with the discharge in a horizontal direction from the truck. Release the hydraulic controls for not less than 2 minutes. Repeat these cycles of depressing the ladder/tower to horizontal and elevating to 75 degrees on headings of 90 degrees and 35 degrees off the heading of the truck, with the discharge rate maintained continuously throughout the movements. Failure of the outriggers or truck tires to remain in contact with the ground, failure of the ladder/tower to maintain elevation and heading within ± 5 degrees of either when the hydraulic controls are released, or bending, twisting, fracture, or failure of any outrigger attachment or component shall constitute failure of this test.

4.5.5 Hydraulic system tests.

4.5.5.1 Hydraulic backup test. Set the outriggers to stabilize the truck. Raise the ladder to maximum recommended elevation and extend the ladder section to the rated condition. With the truck engine off, turn the master electric switch at the control station panel to the "on" position and rotate the ladder 180 degrees. Retract the ladder to the proper bedded position and lower the ladder to the road position. Retract the outriggers to road position. The above process shall require no more than 5 minutes. Restart the truck engine. Failure of the backup pump to complete the test in 5 minutes or failure of the truck to restart shall constitute failure of this test.

4.5.5.2 Pump inlet pressure test. Install a pressure (vacuum) gage in the pump suction line immediately adjacent to the pump and install a thermocouple in the reservoir. Heat the hydraulic fluid to 140 °F ±10 °F. Start the engine and bring it up to maximum governed speed. Operate the water tower while observing the gage and oil temperature. A pressure drop below the pump manufacturer's rating while the hydraulic fluid is at the test temperature and the engine is operating at maximum governed speed shall constitute failure of this test.

4.5.5.3 Hydraulic pump contaminant tolerance. Compare the pump manufacturer's B_{10} contaminant tolerance to the filter B_{10} curve. A contaminant tolerance of the pump that is less than the level of the filter at any point on the B_{10} curve shall constitute failure.

4.5.5.4 Hydraulic filter test. The test may be performed on a hydraulic test stand in place of an on-vehicle test. Connect a pressure gage, temperature gage, and a flowmeter into the filtered (return or pressure) line. Pressurize the line to 100 psig for return-line filters or 1.5 times the maximum system pressure for pressure-line filters. Hold the pressure for not less than 1 minute and then examine for evidence of leaks. Remove the pressure from the line and install another pressure gage on the other side of the filter and a new filter element in the filter housing. Establish a maximum flow through the filter at an oil

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temperature of 150 °F ±5 °F. Maximum flow through the filter shall be the predetermined flow occuring when the boom is depressed while the engine is operating at maximum governed engine speed. Measure and record the pressure drop across the filter assembly. Remove the pressure from the line and plug the filter element chamber. Reestablish and maintain an oil temperature of 150 °F ± 5 °F. Increase upstream pressure to 10 psig above the recorded pressure drop across the filter assembly. Observe the contamination indicator and increase upstream pressure until the indicator reflects that the filter is bypassing. Record the pressure required for bypass opening. Then continue to increase upstream pressure until maximum flow is attained through the bypass valve. Observe and record this pressure. Nonconformance of return-line filters to 3.11.11.1.4.1 or pressure-line filters to 3.11.11.1.4.2 shall constitute failure of this test.

4.5.5.5 Hydraulic cylinder proof test. The test shall be performed on a hydraulic test stand. Secure the cylinder and extend the piston rod to its maximum extended position. Fill the cylinder through the head-end port with oil conforming to MIL-L-2104, grade 10 or MIL-L-46167 and open the rod-end port. Apply pressure through the head-end port with the same oil until the pressure is 1.5 times the pressure relief valve setting. Hold this pressure for not less than 1 minute. Release the pressure and drain the oil from the cylinder while leaving the head-end port open. Retract the piston by filling the cylinder with the same oil through the rod-end port. Apply pressure through the rod-end port with the same oil until the pressure is 1.5 times the pressure relief valve setting. Hold this pressure for not less than 1 minute. Any evidence of rupture, permanent deformation, internal or external damage shall constitute failure of this test.

4.5.5.6 Hydraulic system cleanliness test. Use a liquid automatic particle counter to record particle counts. The particle counter shall be calibrated in accordance with ANSI B93.28 using "AC" Fine Test Dust. Start the truck engine and bring it up to no-load governed speed. Activate the hydraulic pump drive and manipulate all hydraulic controls through full stroke so that each hydraulic circuit has had oil pumped through it for not less than 5 minutes. Connect the calibrated counter into the hydraulic system upstream of the filter(s) used for cleanup or any flow-restricting devices in a return-line having a flow rate which is not less than 5 times the flow rate of the calibrated counter. Connection of the counter shall be through a bypass line and a branch line. The bypass line may return oil directly to the reservoir. The branch line shall tee into the bypass line, shall be connected to the counter on the other end, and shall be not more than 12 inches long. Restart the engine, engage the hydraulic pump drive, and establish a truck engine speed of not less than 1000 rpm. Manipulate the controls through full stroke so that each hydraulic circuit has had oil pumped through it for not less than 5 minutes. Record the maximum particle counts indicated by the counter as each circuit is opened. Any particle count that is greater than the particle counts specified in 3.11.11.1.8 shall constitute failure of this test.

4.5.5.7 Hydraulic external leakage test. Observe and record accumulated leakage at practical intervals over a 12-hour period following all tests involving use of the hydraulic system. Leakage in excess of the limits specified in 3.11.11.1.9 shall constitute failure of this test.

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4.5.5.8 Throttle control switch test. During the test specified in 4.5.4 make minor adjustments to the headings and elevations of the ladder/tower just prior to relinquishing hydraulic control of the boom position. Activate the throttle control switch while making the minor adjustments to headings and elevations. Any surging of the truck engine in response to hydraulic control movements while the throttle control switch is activated shall constitute failure of this test.

4.5.6 Treadplate load test. Unlatch the treadplate at the pump operator's panel and pull it all the way out to the operating position. Load a test deadweight of not less than 400 lbs on the outermost edge of the treadplate. Remove the load and return the treadplate to its retracted position beneath the running board. Unlatch and pull the threadplate all the way out a second time. Failure of the threadplate locking mechanism to latch the treadplate in the retracted position, or binding of the sliding mechanism supporting the treadplate after being subject to the test load, shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Unit of product. For the purpose of inspection, a complete truck prepared for shipment shall be considered a unit of product.

4.6.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.3 Examination. Samples selected in accordance with 4.6.2 shall be examined for the following defects. The AQL shall be 1.0 percent defective.

Defect no.	Defect	Requirement paragraph
118b.	Preservation not as specified for level A, B, or C.	5.1
119b.	Marking not as specified.	5.3

5. PACKAGING

5.1 Preservation. The degree of preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each complete truck, with equipment, shall be preserved in accordance with MIL-STD-281, level A (mobile), with the following additions and exceptions.

5.1.1.1 Engine. Engine and engine accessories shall be preserved in accordance with MIL-E-10062, type I, method I, level A.

5.1.1.2 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-STD-281, except the antifreeze shall conform to MIL-A-46153.

5.1.1.3 Equipment. Equipment shall be secured as provided for, except exposed equipment subject to pilferage, damage, or loss shall be prepared as "on-vehicle equipment" in accordance with MIL-STD-281.

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5.1.2 Level B. Each complete truck shall be preserved in accordance with MIL-STD-281, level B (mobile), and with the additions and exceptions as specified for level A.

5.1.3 Level C. Each complete truck and all equipment issued with the truck shall be suitably preserved to afford protection against deterioration and damage from the contractor to the initial destination. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or desiccated packs. Items having surfaces requiring protection from physical damage, or which are fragile by nature, shall be protected by wrappings, pack compartmentalization, or cartonizing of the individual item. Exposed equipment subject to easy pilferage or loss shall be placed in secured compartments, truck cab, or in wooden boxes. Wooden boxes, when used, shall be secured to the truck with metal strapping and in a manner to prevent damage to the painted surfaces of the truck. The cooling system shall be filled with a fresh clean solution of 50-percent water and 50-percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before damage. The crankcase and all other lubrication reservoirs shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. Crankcase engine oils conforming to MIL-L-21260 are the preferred initial fill. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used and its grade and temperature range. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and other servicing required prior to placing the truck in operation.

5.2 Packing (see 6.7). Each complete truck, preserved as specified in 5.1, shall not require packing.

5.3 Marking (see 6.7). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The truck is intended for use in fighting fires in multistory structures in military operational areas.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. When the truck shall be capable of pumping at altitude, and the service altitude required (see 3.10.1).
d. When hard suction hose shall be furnished (see 3.11.10.7)
e. When EMI emission and susceptibility characteristics are not required (see 3.13).

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f. When exterior markings on the truck are required and the marking data to be used (see 3.18.1).
g. When welding process to be used will be specified by the Government (see 3.19.5.2).
h. Level of preservation required (see 5.1).

6.3 First article. When a first article inspection is required, the item should be a first production model. The first article should consist of one truck. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Definitions.

6.4.1 Ambient temperature. The temperature of the atmosphere measured at the intake side of the engine cooling radiator.

6.4.2 Automatic-type transmission. A transmission in which normal gear ratio changes are effected automatically without manual assist.

6.4.3 Excessive surging. Any movement of the tank contents that would cause the driver of the truck to lose control of the truck during acceleration, deceleration, or any turn.

6.4.4 Flattening. Throughout the bend of a tube, the maximum diameter of the tube shall be equal to or greater than the tube's nominal diameter, and the tube's maximum diameter minus the tube's minimum diameter shall not be greater than 15 percent of the nominal tube diameter.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equip- ment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the U.S. Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-VF, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.7 Packing and marking. Levels of packing have not been included, since they are not appropriate for this item. For purposes of preservation/packing level marking, the marking shall be as required by the customer.

6.8 Subject term (key word) listing.

Fire fighting truck
Fire truck
Truck, fire
Truck, fire fighting

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6.9 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity
Army - ME

Project 4210-A227

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-52942C(ME)	Truck, Fire Tighting: Pumper, 1000 GPM; Ladder/Water Tower Combination

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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